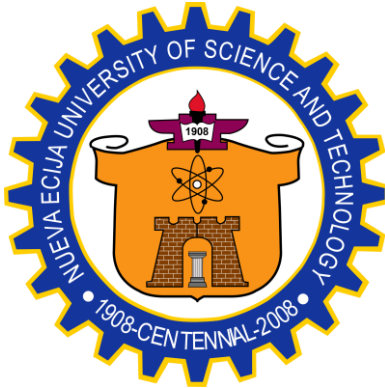


**NEUST GAD PORTAL: A WEB BASED RECORD MANAGEMENT WITH
DECISION SUPPORT FOR THE GAD OFFICE**



A Capstone and Research Project

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Chapter 1

Introduction

Gender and Development (GAD) is a development perspective and process that are participatory and empowering, equitable, sustainable, free from violence, respectful of human rights, supportive of self-determination and actualization of human potential. (CoA, n.d.) It emphasizes participatory, inclusive, and empowering approaches that are free from violence, uphold human rights, and support self-determination and the realization of human potential. At its core, GAD aims to transform societal structures by integrating gender considerations into the design, implementation, monitoring, and evaluation of policies, programs, and projects across all sectors (GAD, 2024; DOST-NCR, 2025).

In the Philippines, the implementation of GAD is guided by key legislation and national frameworks. Executive Order No. 273 approved and adopted the Philippine Plan for Gender-Responsive Development (PPGD) 1995-2025, providing a roadmap of policies and strategies

to ensure that women actively participate in and benefit from national development efforts. Complementing this, the Magna Carta of Women (Republic Act No. 9710) institutionalizes gender mainstreaming as a strategy for promoting women's human rights and eliminating gender-based discrimination in government structures and processes (Philippine Guarantee Corporation, 2025).

Despite these robust legislative frameworks, the operationalization of GAD continues to face several challenges, particularly in the area of record management. Common issues include fragmented data collection practices, the absence of standardized reporting systems, and inadequate technological infrastructure. These obstacles hinder the effective monitoring and evaluation of GAD initiatives, thereby limiting the capacity to generate accurate assessments and develop evidence-based policies (Pagcor, 2025; DOST-NCR, 2025).

Based on interviews conducted with the GAD Focal Person and GAD Director at NEUST, it was revealed that the current paper-based system is time-consuming, error-prone, and

prone to the misplacement of documents. These challenges lead to inefficiencies and delays in tasks such as collecting employee data for events.

In response to these challenges, this study proposes the development of the NEUST GAD Portal—a web-based application designed to streamline the record management processes associated with GAD implementation. This portal will serve as a centralized platform for data collection and storage, enabling more efficient tracking of GAD programs and enhancing transparency, accountability, and informed decision-making in gender and development efforts.

In addition to its core record management functions, the NEUST GAD Portal will incorporate a Research Submission and Approval Workflow. This feature enables faculty and staff to submit GAD-related research proposals and studies directly through the system. Based on interviews each submission will follow a structured review process: first process is initiated by an open call for proposals from the GAD Office. The GAD Focal Person then enrolls interested employees into the system,

who subsequently submit their full proposals through the portal. These submissions are reviewed by the GAD Office, and those that meet the required standards are formally accepted as In-House research projects.

This workflow fosters increased participation in gender-focused research, ensures alignment with institutional priorities and resource availability, and promotes a culture of collaboration, transparency, and systematic evaluation. A similar approach was proposed in the development of an Ethical Clearance workflow system, which addressed common delays caused by manual, paper-based processes in universities. The study highlighted that automating the approval workflow not only improved efficiency but also increased researcher motivation and ensured timely commencement of research activities (Mbabe et al., 2021).

Ultimately, this enhancement supports the academic and operational goals of NEUST's GAD initiatives by streamlining the approval process for gender-related research and development efforts.

Review of Related Literature

This section presents relevant literature and studies that support the development of a web-based GAD Portal, focusing on gender mainstreaming, data management, and ICT integration in public institutions.

Research Repository / Research Submission & Approval Systems

The study by Mbabe et al. (2021) provides a foundational critique of manual, paper-based submission and approval systems, identifying them as a primary source of debilitating delays and a demoralizing factor for researchers. This research is critical as it shifts the focus from the *need* for approval to the *efficiency* of the approval process itself. By developing a workflow system for ethical clearance, the authors demonstrate that automation directly tackles institutional bottlenecks. However, a critical limitation of their work is its narrow focus on the *approval gate* rather than the entire research lifecycle. The system is

designed to manage a compliance hurdle but does not appear to facilitate the subsequent dissemination or utilization of the approved research. This creates a direct link to the GAD Portal, which must not only automate the grading workflow for proposals and finished studies but also ensure that accepted research is seamlessly integrated into a functional repository, thereby adding value beyond the approval stage.

Expanding on the concept of value, the work of Javier et al. (2022) on the InFORMs system offers a critical perspective on the purpose of a repository. It argues that a repository should be a driver for "knowledge-sharing and utilization," not merely a storage vault. Their use of the SCRUM methodology and evaluation against ISO 25010 standards provides a robust framework for ensuring usability, a key lesson for the GAD Portal's development. The critical insight here is that user acceptance is paramount; a technically sound system that is difficult to use will fail as a knowledge-sharing tool. However, a significant gap in their study is the implicit assumption that simply showcasing approved research is sufficient for utilization. The

GAD Portal can critically build upon this by incorporating features that actively promote utilization, such as advanced search filters by GAD theme, keywords, or findings, which the Javier et al. (2022) study does not emphasize.

The repository model proposed by Gholami et al. (2025) introduces a more dynamic and strategic function for a repository: as an active tool for capacity-building and increasing participation. Their "diversity supplement repository" was not passive; it was coupled with mentorship matchmaking and support resources, leading to a measurable doubling in applications. This study is critical because it demonstrates that a well-designed repository can directly influence user behavior and achieve strategic institutional goals. It forces a critique of the Javier et al. (2022) model by showing that a repository can be more than an archive—it can be an intervention. For the GAD Portal, this implies that the repository should not merely store finished GAD research but could also host resources (e.g., sample proposals, GAD frameworks, data analysis tools) to help employees *create better, more impactful GAD*

research in the future, thus creating a virtuous cycle of research and development.

Finally, the national survey by Baek et al. (2024) introduces a crucial, often overlooked dimension: data governance and institutional responsibility. While the other studies focus on efficiency and access, this research critically examines the roles and responsibilities required for compliant and ethical data management. The finding that the roles of "Signing Officials" are poorly defined despite their importance is a stark warning. This directly informs the design of the GAD Portal, which must have clearly defined user roles (e.g., Researcher, Panelist, GAD Focal Person, System Administrator) with explicit permissions. Critically, this study highlights a gap in the others: the lack of discussion on the governance of sensitive research data within a repository. The GAD Portal must therefore be designed not just for usability, but also with a secure, role-based architecture that ensures data integrity and controlled access from the outset.

Records Management / Employee Record Systems

The foundational argument for automation is powerfully made by **Box (2022)** and **Waldron (2024)**, who both critique manual, paper-based systems as being prone to bottlenecks, errors, and inaccessibility. Box (2022) leverages industry survey data from the Association for Information and Image Management (AIIM) to argue that automation provides "improved lifecycle management" and an "enhanced foundation for digital transformation." However, a critical weakness of this source is its commercial nature; it functions as much as a promotional piece for "Dox and Box" as it does an objective analysis, potentially overstating benefits and understating implementation challenges.

Similarly, **Waldron (2024)** effectively highlights the chaos of "disconnected data" in project management, advocating for cloud-based Document Management Systems (DMS) to create a single source of truth. While compelling, this perspective is critiqued for being overly generic, focusing on universal benefits like accessibility without delving into the specific security and governance challenges that such centralized systems

introduce, a gap that subsequent literature addresses.

The theoretical benefits of automation are given empirical weight by **Malak (2025)**, who provides a granular analysis of data automation's role in reducing human error, increasing efficiency, and ensuring data consistency. The citation of a specific 4% error rate for manual data entry powerfully quantifies a problem that other authors only describe qualitatively. Critically, Malak (2025) elevates the discussion by framing automation not as a tool for mere replacement, but for *reallocation* of human capital towards "analysis, decision-making, and management." This is a crucial insight for the GAD Portal, suggesting that its Employee Record Module should not just store data but automate routine tasks (e.g., generating service records, tracking leave balances) to free up HR personnel for more strategic GAD-focused initiatives.

The most significant connection between records management and the core function of an organization is established by Kanyabwira **(2024)**. This study moves beyond operational

efficiency to demonstrate a "significant positive relationship" between electronic records management and HRM effectiveness, including employee motivation and transparent recruitment. This research is critical because it provides an evidence-based justification for the GAD Portal's integrated design. It suggests that a robust Employee Record System is not a peripheral utility but a foundational element that can directly contribute to a positive and transparent work environment, thereby supporting the overall GAD mandate. However, the study's limitation, noted by the author, is its sample size and focus on Rwanda, suggesting a need to contextualize its findings within other cultural and administrative settings, which the GAD Portal will do.

The strategic imperative of managing modern data challenges is explored by Augusta Data Storage (2023), which introduces the concept of "Big Data" and information overload into the records management discourse. This source is critically important for its structured, process-oriented approach, outlining a six-step program from defining objectives to disaster recovery. It correctly

identifies that digitization without a strategy leads to new forms of chaos. However, its primary limitation is its vendor-centric perspective, promoting specific tools and its own services. Despite this, its framework is invaluable, directly informing the need for the GAD Portal to have a clear data classification system, retention policies, and role-based access controls from the outset.

Finally, the critical themes of security and governance are addressed by Data Management Security (2025) and the Indeed Editorial Team (2025). These sources reframe records management from an efficiency project to a risk management and compliance obligation. Data Management Security (2025) starkly outlines the financial and reputational costs of data breaches, emphasizing the need for a "defense-in-depth" strategy. This serves as a critical counterbalance to the open-access ideals of cloud-based DMS promoted by Waldron (2024), forcing a synthesis that prioritizes both accessibility and security. The **Indeed Editorial Team (2025)** further reinforces this by defining records management as essential

for "compliance, risk management and governance," highlighting the special sensitivity of employee records containing personally identifiable information (PII). This directly mandates that the GAD Portal's Employee Record Module must be designed with stringent security protocols that exceed those of the public Research Repository.

GAD (Gender and Development) Policies / Guidelines

The policy landscape is well-defined by strategic frameworks from major institutions. The Asian Development Bank (ADB, 2025) and the Ministry of Education, Youth and Sport in Cambodia (MOEYS, 2023) exemplify the global commitment to GAD, with ambitious targets and detailed mainstreaming plans. Similarly, the Philippine Commission on Women (PCW, n.d.) operationalizes this through the National GAD Resource Program (NGRP), focusing on capacity building and technical assistance. These documents establish GAD as a non-negotiable component of modern development and public administration. However, a critical analysis reveals that these are largely *aspirational* and *input-*

focused documents. They state what should be done but offer limited insight into the mechanics of tracking whether these plans lead to meaningful change, a gap that empirical research forcefully exposes.

The study by Duma (2022) provides a critical bridge between policy and practice within an educational context directly relevant to GAD Portal. The finding that gender perspectives are "less mainstreamed into the policies" due to a "lack of gender analysis and gender mainstreaming plan" is a stark indictment of implementation failure. This research is crucial because it identifies the specific, on-the-ground barriers that high-level policies like MOEYS's cannot overcome alone. It directly justifies the need for the GAD Portal's research repository; by centralizing GAD research and successful plans, the portal can serve as a readily accessible knowledge base to combat the very "insufficient knowledge and training" that Duma (2022) identifies as a key hindering factor.

The most severe critique of current GAD paradigms comes from Gupta et al. (2023), who

argue that gender mainstreaming has largely failed to deliver transformative results. Their finding that mainstreaming has been limited to "altering internal processes rather than emphasizing results" is a powerful critique of a tick-box compliance culture. They compellingly link this failure to institutional cultures that are "hierarchical" and driven by "short-term crisis response." This study forces a critical evaluation of any new GAD initiative: will it merely automate existing, flawed processes, or will it drive genuine change? The GAD Portal must be designed to answer this challenge by being inherently results-focused. Its grading system for research proposals can prioritize studies with clear, transformative gender equality outcomes, and its repository can track and showcase the real-world impact of completed GAD research, moving beyond simple archival.

The empirical evidence for the failure of mainstreaming is further solidified by Ose et al. (2025), whose longitudinal study in Norway found no increase in research with a gender perspective despite a decade of gender mainstreaming policies. This is a critical

piece of evidence that shifts the discussion from anecdotal to data-driven. The authors attribute this to "symbolic policy implementation," where policies exist on paper but are not effectively enforced or integrated. This finding has a direct and critical implication for the GAD Portal: it must be more than a passive repository. To avoid being another symbol of compliance, it must have active, embedded mechanisms—like mandatory gender analysis frameworks in the research submission process and panels trained to evaluate gender perspectives—that *compel* the integration of GAD into the institutional research culture.

Finally, the scoping review by Condrón et al. (2024) on gender equality training in higher education reveals a field "lacking coherence," with vague theoretical underpinnings and poor evaluation of long-term impact. This critiques the capacity-building component that is central to GAD policies (like the PCW's NGRP). It suggests that training alone is insufficient without a structured system to reinforce and apply learning. The GAD Portal can critically address this by not only hosting training

materials but by providing a platform where students and employees *apply* their learning by developing and submitting GAD-focused research, thereby creating a direct link between knowledge and practice.

Impact of GAD Initiatives and the Role of Monitoring & Evaluation

Monitoring and evaluation (M&E) are vital mechanisms that determine the success of Gender and Development (GAD) initiatives in transforming institutional practices and promoting gender equality. In the Philippine context, GAD policies have evolved to incorporate evidence-based monitoring frameworks that enhance accountability and inclusiveness across all government levels. The Philippine Statistics Authority (2021), through the *Approval and Adoption of the Updated Philippine Core Gender and Development Indicators*, reinforced the importance of standardized measurement tools to track gender outcomes. The updated indicators align local monitoring efforts with international frameworks such as the United Nations Sustainable Development Goals (SDGs) and the Asia-Pacific Economic Cooperation (APEC)

Women and Economy Dashboard. Critically, while this move ensures statistical consistency and comparability, the PSA's report also underscores persistent data gaps and capacity constraints among implementing agencies. These gaps often translate into fragmented data collection practices and delayed reporting—issues that weaken the empirical basis for evaluating GAD programs. Thus, while the indicators establish a strong policy foundation, their success hinges on the presence of efficient record management systems and digital infrastructures that can support timely data generation.

In a broader national perspective, the Gender Country Profile for the Philippines (2021) situates GAD monitoring within global benchmarks and post-pandemic realities. The report acknowledges the Philippines' sustained ranking among the top gender-equal countries globally but warns that progress has slowed due to uneven implementation and weak institutional evaluation mechanisms. The COVID-19 pandemic further exposed these deficiencies, as women's economic participation and access to reproductive health services deteriorated amid limited

monitoring of gender-based vulnerabilities. The Profile emphasizes the need for intersectional, data-driven analysis to guide gender-sensitive recovery strategies, illustrating how M&E systems should adapt to emerging social contexts rather than function as static reporting tools. This insight highlights the urgency of digital transformation in GAD tracking systems to ensure real-time and responsive evaluations—an area that your proposed NEUST GAD Portal seeks to address.

Empirical studies such as Moreno (2023) expand the discussion by evaluating the impact of gender-responsive budgeting (GRB) in Region IX local government units. Moreno's comparative analysis revealed that the degree of M&E integration directly influences the effectiveness of GRB implementation. LGUs with strong monitoring mechanisms demonstrated measurable improvements in gender equity outcomes, while those with weak evaluation systems exhibited poor compliance and data inconsistencies. Importantly, Moreno argues that successful M&E requires not only political will but also a clear technological and procedural infrastructure. His

recommendation to establish standardized manuals and digitalized reporting channels supports the notion that efficient monitoring is both a technical and governance issue. This aligns with the need for higher education institutions (HEIs) like NEUST to institutionalize M&E frameworks within their GAD systems to ensure accuracy, transparency, and participatory evaluation.

Complementing these perspectives, the Philippine Commission on Women's (PCW, 2022) *Updated Gender Equality and Women's Empowerment (GEWE) Plan 2019-2025* underscores M&E as a strategic pillar in mainstreaming gender concerns across agencies. The GEWE Plan integrates monitoring mechanisms into policy implementation to evaluate the inclusiveness and responsiveness of government programs, especially during crises such as the COVID-19 pandemic. However, the report also concedes that limited technical expertise and the absence of automated monitoring platforms restrict the plan's full operationalization. This observation mirrors the common institutional challenge where M&E remains a compliance requirement rather than a transformative management tool. Critically,

the PCW's emphasis on aligning local GAD plans with national evaluation frameworks points to the necessity of a harmonized, digital, and participatory approach—elements central to the proposed NEUST GAD Portal's design.

Finally, in the higher education sector, Palma (2024) provides micro-level insights into the impact of GAD implementation and awareness within a state university. Her study found that awareness, satisfaction, and implementation of GAD initiatives were interlinked, with awareness significantly influencing perceived program effectiveness. Palma identified moderate implementation and limited satisfaction among faculty and students, citing inadequate monitoring and lack of systematic evaluation as key barriers. Her recommendation for a *GAD Capability Enhancement Framework* underscores the need to strengthen both technical and evaluative capacities within institutions—a finding that reinforces the premise that digital monitoring systems can improve data integrity, engagement, and overall institutional accountability.

Taken collectively, these studies reveal a consistent pattern: while the Philippines has developed robust legislative and policy frameworks for gender equality, the impact of GAD initiatives remains constrained by weak monitoring, fragmented data management, and limited digital integration. The adoption of standardized indicators (PSA, 2021), strategic frameworks (PCW, 2022), and localized evaluation studies (Moreno, 2023; Palma, 2024) all point toward the same conclusion—that monitoring and evaluation are not peripheral activities but central mechanisms for ensuring institutional effectiveness and policy coherence. The convergence of these findings provides a strong justification for the development of an integrated digital M&E platform, such as the NEUST GAD Portal, to enable real-time tracking, informed decision-making, and sustainable gender mainstreaming within higher education.

Gender Mainstreaming and Institutional Capacity Building

The effectiveness of gender mainstreaming initiatives depends largely on

an institution's ability to strengthen its internal capacity, integrate gender perspectives into organizational structures, and establish evidence-based mechanisms for sustained implementation. The ESCAP Progress Report on Gender Mainstreaming (2023) emphasizes that institutional capacity building is essential in mitigating the gendered impacts of global crises such as the COVID-19 pandemic. ESCAP's policy direction from 2023-2027 highlights the need for gender-responsive programming and continuous staff training as part of post-pandemic recovery efforts. Notably, ESCAP demonstrates leadership by achieving gender parity across all organizational levels, underscoring that institutional transformation must begin internally before extending to policy and program design (United Nations Economic and Social Commission for Asia and the Pacific [ESCAP], 2023). This aligns with the principle that gender mainstreaming cannot be sustained solely through advocacy; it requires systemic reforms and capability development embedded within institutional frameworks.

Building on this institutional perspective, Palma (2024) examined the

mediating relationships among awareness, implementation, and satisfaction in a public higher education institution (HEI) in the Philippines. Her study found that while respondents exhibited moderate awareness and satisfaction levels, the “people” component—reflecting the role of individuals in executing gender policies—had the strongest influence on implementation outcomes. The findings underscore that gender mainstreaming in HEIs is not merely procedural compliance but an evolving process shaped by stakeholder engagement and institutional learning. Palma’s proposed *Gender and Development Capability Enhancement Framework* provide a model for integrating awareness-raising, training, and evaluation into institutional systems, demonstrating that digital tools and structured frameworks can enhance GAD implementation effectiveness.

Similarly, Rañon (2024) conducted an assessment of gender mainstreaming at Camarines Sur Polytechnic Colleges using the Enhanced Gender Mainstreaming Evaluation Framework (E-GMEF). The institution achieved Level 3 under GAD application, indicating substantial progress in policy development and

people involvement but revealing weaker scores in programs, projects, and enabling mechanisms. The study identified the need for strengthened training, policy integration, and gender-responsive planning to elevate institutional capacity. Rañon's findings emphasize that the sustainability of gender mainstreaming depends on continuous organizational learning, supported by formal monitoring frameworks such as E-GMEF that can quantify institutional progress and guide strategic interventions.

Expanding on these institutional case studies, Perez et al. (2023) offered a systematic review of 130 studies on gender mainstreaming practices in HEIs. Their review revealed that while strong policy frameworks and leadership support are key facilitators, persistent barriers—including limited expertise, weak monitoring systems, and lack of gender-disaggregated data—undermine the long-term institutionalization of GAD. Importantly, the review situates these challenges within the Philippine context, where only a minority of HEIs maintain functional GAD offices or utilize gender-sensitive databases. The authors call for

enhanced accountability mechanisms, resource allocation, and capacity-building initiatives that integrate digital monitoring systems and sustained leadership engagement. Their findings reinforce the need for institutional portals and knowledge platforms that consolidate GAD data, support continuous training, and provide transparent evaluation metrics.

Collectively, these studies demonstrate that gender mainstreaming and institutional capacity building are mutually reinforcing processes. Effective mainstreaming requires not only compliance with policy mandates but also the establishment of robust institutional mechanisms—training systems, monitoring frameworks, and digital infrastructures—that enable adaptive learning and accountability. For institutions such as NEUST, embedding these capacity-building principles into digital GAD portals can operationalize mainstreaming efforts, align with national and international gender equity frameworks, and strengthen institutional readiness to address gender issues in a rapidly changing environment.

IT Systems / Software Development / Change Management

The foundational user-centric challenge is articulated by **Singh et al. (2025)**, whose systematic review on the digital divide critically identifies a "lack of digital literacy and skills" as the primary barrier at the *usage* level. This is not merely a problem of access but of effective adoption. This finding is directly operationalized by **Najjar (2024)**, who argues that "user education is a fundamental aspect of effective IT support." While Najjar's perspective is practical, focusing on reducing support tickets, it critically complements Singh et al. by providing a solution: "comprehensive training programs" and a "well-organized knowledge base." For the GAD Portal, this synthesis mandates that a robust training and support strategy is not optional but essential to ensure it does not inadvertently worsen the gender digital divide by sidelining less tech-savvy users.

However, training alone is insufficient if the organizational culture is hostile to change. The **Panorama Consulting Group (2024)** provides a critical analysis of this, identifying "cultural resistance" as a primary cause of IT project failure. Their insight that resistance stems from "fear of the unknown," "mistrust in leadership," and "cultural attachment to legacy systems" moves the discussion beyond individual skill gaps to systemic organizational dynamics. The recommended strategy of Organizational Change Management (OCM), including tailored communication and cultivating "a culture of change," is crucial. This forces a critical evaluation of the GAD Portal's rollout: it must be framed not as a top-down mandate, but as a collaborative tool for empowerment, directly addressing the "fear of the unknown" by clearly articulating its benefits to employees' daily work.

Even with a supportive culture and good training, a poorly designed system can fail. **Shahrzadi et al. (2024)** introduce the critical concept of "information overload," noting its negative effects on "decision-making" and "productivity." Their finding that overload is caused by "information

characteristics" and "information technology parameters" is a direct design warning for the GAD Portal. The repository must not become a digital dumping ground; it requires intelligent "filtering and prioritizing" mechanisms, intuitive information architecture, and clean user interfaces to ensure it is a tool for insight, not confusion.

The empirical study by **Belas et al. (2025)** adds a vital gendered lens to these challenges, finding "significant differences" in how men and women perceive the value of digitalization. This evidence critically demonstrates that technology adoption is not uniform. A one-size-fits-all training or communication strategy would be inadequate. The portal's value proposition and functionality must be designed to resonate with these diverse perspectives, for instance, by highlighting its role as a knowledge repository and risk mitigation tool, which their study found particularly valued by female entrepreneurs.

At the strategic level, the UNIDO report by **Corneliussen et al. (2023)** and the study on educational technology by **Berezhna et al. (2025)** provide the high-level mandate and a

cautionary note, respectively. Corneliussen et al. warn that without "enabling policies," technology can exacerbate exclusion, while Berezhna et al. note the "significant challenges" in creating effective digital learning tools. These sources collectively argue that the GAD Portal must be a deliberately inclusive and pedagogically sound platform.

Finally, the technology stack itself is a critical consideration. Vidjikan & Neville (2025) present a balanced view of PHP, acknowledging its maturity and vast ecosystem but also its "decline in popularity" and perception as less "cutting-edge." A critical reading suggests that while PHP is a viable choice, the decision carries implications for developer recruitment, long-term maintainability, and the ability to implement modern, reactive user interfaces—a key factor in combating the information overload identified by Shahrzadi et al.

Data-Driven / Decision Making / Technology in Management

The foundational argument for this shift is powerfully made by **Skan AI Contributors (2024)**, which positions data-driven decision making (DDDM) as an antidote to the "costly and regrettable mistakes" of intuitive leadership. The article effectively outlines key benefits, such as increased accountability, continuous improvement, and competitive advantage, by framing data as an objective foundation for strategic choices. However, a critical reading exposes this perspective as somewhat idealized. It understates the significant organizational inertia and "resistance to change" that it itself identifies as a barrier. The source's commercial nature, promoting its own "AI-powered technologies" and "Process Intelligence," also suggests a potential bias towards technological solutionism, potentially oversimplifying the profound cultural shift required to become truly data-driven.

The broader economic and industrial impact of this technological shift is explored by ResearchFDI (2023). This article critically examines how technology acts as a "catalyst for change," driving automation,

digitalization, and the creation of a new digital economy. Its analysis of "job displacement" alongside new opportunities for "reskilling and upskilling" presents a balanced view of technological disruption. A crucial insight is the "increasing importance of data and analytics in decision-making," which directly links to the DDDM philosophy. However, the article's critical contribution lies in its discussion of the challenges, particularly the "digital divide." It correctly identifies that unequal access to technology and digital literacy can worsen existing inequalities, creating a stark contradiction: the very tools meant to drive inclusive economic development can, if mismanaged, become engines of exclusion.

Synthesis

A critical synthesis of the literature across the domains of research management, records systems, GAD policy, IT implementation, and data-driven management reveals a consistent and compelling narrative: the failure of disparate, manual systems to support the strategic goals of Gender and Development. The literature collectively identifies a critical

implementation gap where high-level GAD policy ambitions are consistently undermined by inefficient workflows, inaccessible knowledge, and a lack of actionable data. The GAD Portal emerges as a direct response to this gap, positioned not as a mere technological tool but as an integrated platform that synergizes four critical pillars to transform GAD from a theoretical commitment into an operational reality.

The foundational need for the portal is established by the documented inefficiencies in research management and records systems. Manual, paper-based processes for research approval and employee records create debilitating bottlenecks, errors, and a lack of transparency, ultimately demoralizing researchers and hindering strategic human resource management (Mbabe et al., 2021; Kanyabwira, 2024). This operational failure is compounded by the well-documented shortcomings in GAD implementation, where policies remain aspirational due to a lack of practical tools, a focus on compliance over impact, and insufficient capacity building, leading to "symbolic" rather than transformative gender mainstreaming (Gupta et al., 2023; Ose et al., 2025; Duma, 2022). The

GAD Portal directly addresses these dual failures by automating workflows to ensure efficiency and embedding GAD principles—through mandatory gender analysis frameworks and impact-focused grading criteria—directly into the institutional research lifecycle. However, building a functionally sound system is insufficient without ensuring its adoption and effective use. The literature on IT systems and change management provides a crucial caution that technology implementation is a socio-technical challenge. The success of the GAD Portal is contingent upon proactively managing cultural resistance (Panorama Consulting Group, 2024), overcoming the digital divide through inclusive design and comprehensive training (Singh et al., 2025; Najjar, 2024), and designing against information overload with intuitive interfaces and smart filtering (Shahrzadi et al., 2024). Furthermore, the system must be architected with a gendered lens, recognizing that men and women may perceive and value digital tools differently, to ensure it is embraced as an empowering resource by all employees (Belas et al., 2025).

Ultimately, the portal's highest value is realized by evolving into a nerve center for Equity-Focused, Data-Driven Decision Making (Skan AI Contributors, 2024; ResearchFDI, 2023). By consolidating data on research projects, employee participation, and training outcomes, the portal transforms scattered information into actionable intelligence. This allows leadership to move beyond intuition, making strategic decisions about resource allocation, capacity-building needs, and policy direction based on empirical evidence of what truly advances gender equality within the organization. This data-driven approach is securely underpinned by robust governance frameworks that protect sensitive information, ensuring the system's integrity and compliance (Baek et al., 2024; Data Management Security, 2025).

In conclusion, the GAD Portal is conceptualized as a synergistic solution that integrates the key insights from the literature into a coherent whole. It applies automation to solve foundational inefficiencies in research and records management; it uses structured workflows and intelligent repositories to operationalize GAD policies and ensure meaningful, impact-

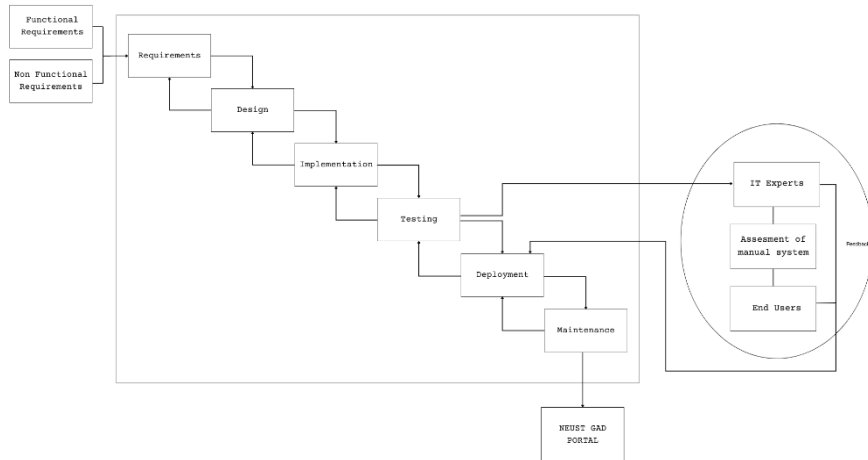
oriented implementation; it leverages principles of change management and inclusive design to guarantee widespread adoption and use; and it employs **data** analytics and governance to provide the insights necessary for strategic, evidence-based GAD planning. By uniting these four pillars, the GAD Portal transcends its function as a simple IT system and becomes the foundational infrastructure for building a more efficient, transparent, accountable, and genuinely gender-responsive organization.

Conceptual Framework

The conceptual framework of this study follows the Modified Waterfall Software Development Life Cycle (SDLC) model to guide the creation of a Web-Based Record Management System with Decision Support for the university GAD Office. The Modified Waterfall model ensures a structured and systematic approach to software development, allowing for a clear and logical progression through distinct phases such as planning, analysis, design, implementation, testing, deployment,

and maintenance. Figure 1 below shows the paradigm of the study.

Figure 1. **Research Paradigm**



As shown above, the development process begins with the identification of both functional and non-functional requirements. Functional requirements define the core features and tasks the system must perform, while non-functional requirements address system attributes such as security, usability, reliability, and scalability. These requirements serve as the foundation for the Requirements phase, where all system needs are carefully documented and validated with stakeholders to ensure alignment with the objectives of the NEUST GAD Office.

Following this, the design phase is initiated. In this stage, the researchers translate the gathered requirements into a detailed blueprint of the system, including the system architecture, user interface layout, and database schema. This structured design provides the technical foundation necessary to guide accurate development and integration of features within the Web-Based Record Management System with Decision Support.

Once the design is finalized, the project proceeds to the Implementation phase. Here, developers carry out the actual coding process—creating the system's backend functionalities, frontend interface, and database modules. This phase also involves system integration to ensure all parts of the application work harmoniously. After implementation, the system undergoes thorough Testing to detect and resolve any bugs, performance bottlenecks, or functionality gaps. This guarantees that the final product aligns with the documented requirements and is ready for deployment.

The Deployment phase involves rolling out the developed system to its intended users. IT

Experts and End Users participate in the evaluation process. IT Experts review the system's technical performance, security, and stability, while End Users assess its usability, comparing it with the current manual system. This evaluation also includes a structured assessment of the manual system, enabling a direct comparison to validate improvements brought by automation.

Subsequently, the Maintenance phase ensures long-term system performance and user satisfaction. This includes applying system updates, addressing security vulnerabilities, enhancing system features based on evolving user needs, and continuously integrating feedback from both technical evaluators and everyday users.

This conceptual framework reflects the linear, structured approach of the Modified Waterfall SDLC model, emphasizing a clear sequence of development stages with defined deliverables. By integrating user assessment and expert feedback at key points—especially during deployment and maintenance—the framework supports the creation of a dependable, efficient, and user-centered GAD Portal. Ultimately, the system aims to enhance

record management processes and support data-driven decisions within the GAD Office.

Statement of the Problem

This study aims to develop and evaluate a system based on various constructs and software characteristics. Specifically, it seeks to answer the following research questions:

1. How may the manual system be described on the following constructs:
 - 1.1 security of files and records;
 - 1.2 management of data;
 - 1.3 reliability of output;
 - 1.4 efficiency of processes; and
 - 1.5 compliance with regulations and standards?
2. How may the proposed project be described based on the stages of the Modified Waterfall SDLC Model:
 - 2.1 planning and requirements gathering;
 - 2.2 design and prototyping;
 - 2.3 development and implementation; and
 - 2.4 testing and deployment?
3. How may the proposed project be evaluated by the elected IT experts following the

qualifications of software characteristics standards which include:

- 3.1 functional suitability;
- 3.2 performance efficiency;
- 3.3 compatibility;
- 3.4 usability;
- 3.5 Security;
- 3.6 Maintainability;
- 3.7 Potability?

4 How may the candidate end-users assess the proposed project's standard concerning the selected software characteristics standards which include:

- 4.1 functional suitability;
- 4.2 performance efficiency;
- 4.3 usability;
- 4.4 reliability; and
- 4.5 security?

Scope and Delimitation

The Web-Based Gender and Development (GAD) Portal is a comprehensive system developed for implementation across selected campuses of the Nueva Ecija University of Science and Technology (NEUST). These include General Tinio Street Campus, Sumacab Campus,

San Isidro Campus, Gabaldon Campus, Fort Magsaysay Campus, and Atate Campus—campuses where officially recognized GAD offices are present. The system will not be implemented in off-campus or non-NEUST institutions, as these campuses are managed externally and do not fall under the administrative jurisdiction of NEUST.

The system aims to streamline the management of gender-related employee data and strengthen the operational capacity of the university's GAD offices. It includes core functionalities such as centralized employee data storage, inventory report generation, and role-based account management.

A key feature of the GAD Portal system is its integrated Research and Study Submission module, which enables employees to submit gender-focused research proposals directly through the platform. Each submission is initially reviewed by the GAD Focal Person of the respective department to ensure alignment with gender and development objectives. Once validated, the proposal is endorsed to the RET Chair for further evaluation and announcement of open research submissions, and subsequently

assessed by the assigned panel members. The evaluation process includes commenting and voting on research proposals, with all feedback and decisions systematically recorded for transparency.

The system supports six primary user roles: the Director, Secretary, GAD Focal Person, Employee, RET Chair, and Panel. The Director oversees all portal operations, including viewing system dashboards, managing accounts, and accessing logs of all user activities. The Technical Assistant assists in managing inventories, generating reports, and creating accounts for GAD Focal Persons. The GAD Focal Person handles employee record management, generates employee reports, and facilitates research submissions from employees. Employees can update their profiles, submit research studies, and tag co-authors. The RET Chair manages research announcements and oversees report generation for approved studies. Finally, the Panel members evaluate submitted research through comments and voting mechanisms. This structured role distribution ensures smooth workflow, accountability, and consistent

monitoring of gender-related research initiatives within the institution.

The system is exclusively designed for managing records of NEUST employees. It will not support student data management or be used by external organizations. Its implementation is strictly limited to main campuses and will not be extended to other NEUST departments that are unrelated to GAD functions. The focus remains on creating a centralized and inclusive platform that supports efficient data handling, encourages employee participation in gender initiatives, and ensures compliance with national GAD frameworks.

In conclusion, the scope of the study ensures that the GAD Portal effectively addresses the data management and research support needs of NEUST's GAD offices. The delimitations clarify that the system is confined to NEUST employees within designated campuses and does not include students, external institutions, or non-GAD university departments.

Significance of the Study

The development of the Gender Development Support Management System for the GAD office aims to improve gender-related decision-making, employee management, and administrative efficiency through data-driven decision-making (DDDM). The system will primarily benefit faculty members and administrators, but its impact will extend to various stakeholders within the institution. The following groups will gain significant advantages from the successful implementation of this project, arranged in order of who will benefit the most:

GAD Office. As the primary user of the system, the GAD office will experience significant improvements in managing gender-related data, retirement tracking, and gender-specific resource distribution. The system will enhance decision-making through real-time data visualization and predictive insights, allowing GAD administrators to allocate resources effectively, monitor gender statistics, and ensure fair and equitable distribution of support initiatives. The automation of tasks such as

notification of item receipt and employee confirmation will reduce administrative workload and improve efficiency.

GAD Focal Persons. The focal person of NEUST, as the direct subjects of gender-related policies and programs, will greatly benefit from the system. By ensuring transparent and data-driven decision-making, the system promotes fair distribution of gender-sensitive benefits and better planning for retirement transitions. Faculty members will also receive timely notifications about available resources, making it easier for them to access support from the GAD office. Additionally, the system's ability to track gender demographics ensures that faculty needs are addressed equitably.

Administration. The university administration will benefit from the system's ability to provide comprehensive workforce analytics, including retirement forecasting and gender distribution data. This will help in formulating institutional policies related to workforce planning, hiring strategies, and diversity initiatives. The system will also contribute to greater compliance with gender-related policies, ensuring that the university

fosters an inclusive and well-supported faculty workforce.

Students. Although the current implementation is limited to faculty members, future expansions of the system could extend its benefits to students by promoting gender inclusivity and awareness within the institution. The data-driven insights provided by the system could be utilized to improve gender-related programs, student support services, and policies that foster inclusivity in education. While students are not direct beneficiaries at this stage, the long-term impact of a more inclusive faculty and institutional policies will ultimately enhance the learning environment at the university.

Gender Advocacy Groups and Policymakers.

Gender advocacy organizations and policymakers focused on workplace diversity and gender equality may also benefit from the system. The system's real-time gender analytics and demographic data could provide valuable information for research and policymaking efforts. Additionally, as the university strengthens its gender development programs, the system could serve as a reference model for other institutions looking

to implement data-driven gender policies and decision-support systems.

Definition of Terms

This section defines key terms to ensure clarity and consistency throughout the study. These definitions establish a foundation for understanding the implementation of the Web-Based Record Management System with Decision Support for the GAD Office, particularly in the context of gender development, employee management, and decision-making.

Age Tracking. A system feature that monitors employees nearing retirement and ensures proper recognition and workforce transition planning.

Data-Driven Decision Making (DDDM). Is the process of collecting and analyzing data to guide decision-making processes. This approach involves gathering data based on key performance indicators (KPIs) and transforming that data into actionable insights to optimize operations and enhance decision-making accuracy. (Asana, 2024)

Decision Support System (DSS). A technology-driven framework that assists organizations in making informed choices based on data analysis.

Employee Management System (EMS). A digital system that maintains employee records, automates HR processes, and provides data for strategic decision-making.

Gender and Development (GAD). A development perspective and process that is participatory and empowering, equitable, sustainable, free from violence, respectful of human rights, supportive of self-determination, and actualization of human potentials. It seeks to achieve gender equality as a fundamental value that should be reflected in development choices. (GSIS, n.d.)

Gender Identity. A person's inner sense of being a boy/man/male, girl/woman/female, another gender, or no gender. It is how one self-identifies as a result of a combination of inherent and extrinsic or environmental factors. Gender identity is not the same as sexual orientation. (*Terminology*, 2024)

Gender Mainstreaming. The process of integrating gender perspectives into any planned action, including legislation, policies, or programs, in all areas and at all levels. (*Handbook on Gender Mainstreaming for Gender Equality Results*, 2024)

Gender Resource Hub. A centralized platform for knowledge-sharing, research, and best practices related to gender issues and development. (CGIAR, 2021)

Gender Sensitivity. The ability to acknowledge and highlight existing gender differences, issues, and inequalities and incorporate these into strategies and actions. (INEE, n.d.)

Information and Communication Technology (ICT). ICT encompasses a diverse set of technological resources, including computers, the internet, and software applications, that facilitate the creation, storage, retrieval, and analysis of data. (UNESCO IIEP Learning Portal, n.d.)

Issuance Inventory Management. A feature within the system that organizes and tracks gender-sensitive gifts and benefits for employees.

LGBTQ Inclusion. Policies and practices that ensure fair treatment and support for individuals of diverse sexual orientations and gender identities. (Turner, 2025)

Software Development Life Cycle (SDLC).

A systematic process used in software development that includes planning, analysis, design, implementation, testing, deployment, and maintenance. (Jackson, 2023)

System Usability Scale (SUS). A method used to evaluate the ease of use, efficiency, and user satisfaction of a software application. (*We Are Testers*, 2025)

Web-Based Record Management System. An online platform designed to store, manage, and analyze records for improved efficiency and decision-making.

Workforce Demographics. The statistical analysis of an organization's employee composition, including gender, age, and other key factors, to support HR planning.

Chapter 2

METHODOLOGY

This research section presents the systematic process for the successful development and employment of the study. It includes the research design, locale, respondents, sampling technique, research instruments, data gathering procedures, data analysis technique, and ethical considerations.

Research Design

The research design serves as the blueprint for conducting the study, outlining the overall approach used to collect, analyze, and interpret data (Maxwell, 2022). This study employs a descriptive-developmental research design, which combines the systematic documentation of current conditions with the development and evaluation of a prototype system. This approach is suitable for studies aiming to design, develop, and assess the effectiveness of a technological solution based on existing needs and practices.

The descriptive aspect of this study focuses on identifying and documenting the current experiences, challenges, and needs of the related stakeholders in implementing

gender development initiatives. Using surveys and interviews, the research will gather detailed feedback on how gender-related data is currently managed and what improvements are needed to support decision-making processes more effectively.

The developmental aspect involves the design and creation of a prototype system—specifically, the NEUST GAD Portal. This system will be developed in response to the identified needs and will undergo preliminary testing with selected users. Feedback from this testing phase will be used to refine the system's functionality, interface, and data reporting capabilities to ensure its relevance and usability.

To ensure comprehensive and accurate data collection, the study will employ total enumeration as its primary method. All designated GAD Focal Persons from each department across various NEUST campuses will be included as respondents, along with the GAD Director, Secretary and RET Chair Person. This approach ensures that all departments' perspectives are represented without the need

for sampling, enhancing the depth and reliability of the findings.

Furthermore, the research will incorporate prototype testing and evaluation to assess user satisfaction, ease of use, and the system's potential impact on gender data management and policy formulation. This will provide essential feedback that guides system refinement prior to wider implementation.

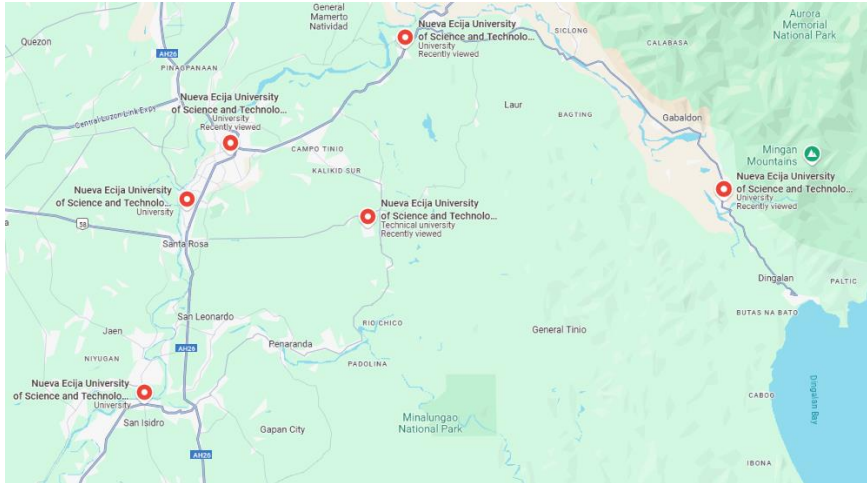
By employing a descriptive-developmental approach, supported by total enumeration and prototype validation, this research ensures a systematic and practical path toward creating a user-centered and data-informed Gender Development Support System for NEUST faculty members.

Research Locale

In this study, the research locale is Nueva Ecija University of Science and Technology (NEUST), specifically focusing on campuses with established Gender and Development (GAD) offices: General Tinio Street Campus, Sumacab Campus, San Isidro Campus, Gabaldon Campus, Atate Campus, and

Fort Magsaysay Campus. Figure 2 below shows the locator map.

Figure 2. **Locator Map**



(Source: Google Maps. (2025). [Map of Nueva Ecija, Philippines]. Retrieved from <https://www.google.com/maps/>)

The selection of the Nueva Ecija University of Science and Technology (NEUST) as the research locale is grounded in the university's recognized commitment to institutional advancement and gender-responsive governance. From the outset, the study was designed to focus on campuses directly managed and operated by NEUST, ensuring consistent administrative structures and reliable implementation of system features.

The study covers the following NEUST campuses: General Tinio Street Campus, Sumacab Campus, San Isidro Campus, Gabaldon Campus, Atate Campus, and Fort Magsaysay Campus. These campuses were deliberately chosen not based solely on the presence of an active GAD office, but because they fall under the official jurisdiction and management of NEUST. This approach ensures that the developed Web-Based Record Management System with Decision Support can be integrated seamlessly across campuses governed by the university, regardless of the current operational status of their GAD offices.

Exclusions were made for off-campus locations such as Talavera Campus, which, although affiliated, are independently managed and therefore outside the scope of NEUST's internal administrative systems. This decision allows the study to remain focused on improving internal processes and ensuring that the system aligns with the university's existing frameworks and long-term development goals.

By including NEUST-administered campuses with diverse academic and administrative environments, the study provides a comprehensive foundation for evaluating the system's relevance, adaptability, and effectiveness. It also ensures that data collected from different stakeholders—such as employees, GAD Focal Persons, and administrators—will reflect the actual needs and workflows within the university.

Research Respondents

The respondents of this study are individuals who will provide essential insights for the development and evaluation of the NEUST Gender and Development (GAD) Portal. To ensure that the data collected are meaningful and aligned with the study's objectives, a purposive sampling technique will be employed.

Purposive sampling will be used to identify key personnel who play vital roles in the implementation and oversight of GAD operations within the university. These include one (1) GAD Director, one (1) GAD Secretary or Technical Assistant, thirty-three (33) GAD Focal Persons representing each department or

campus, and one (1) RET Chairperson. These respondents were chosen based on their direct responsibilities in managing gender-related employee data, overseeing GAD programs, and making strategic and administrative decisions aligned with institutional and national gender policies.

Moreover, 30 IT experts will be consulted to evaluate the system's technical feasibility, security, and overall performance. These experts will provide feedback on the system's design, scalability, and potential improvements, ensuring that the web-based record management system with decision support meets industry standards. To ensure the selection of qualified evaluators, purposive sampling will be used. The chosen IT experts will meet the following criteria:

Educational Background. Must have a degree in Information Technology, Computer Science, or a related field.

Professional Experience. Should have at least one year of experience in software or

systems development, system evaluation, or academic instruction in IT-related subjects.

Relevant Expertise: Preferably knowledgeable in web-based systems, records management systems, or decision support systems.

Availability: Must be available and willing to actively participate in the evaluation process, providing thorough and constructive feedback.

This sampling method ensures that the selected IT experts possess the necessary skills and knowledge to provide valuable insights into the system's functionality and performance.

The final total number of respondents will be determined based on the total number of departments across all NEUST campuses, ensuring that the collected data is comprehensive and reflective of the institution's diverse academic structure. The selection criteria for respondents are based on their role within the university, their involvement in gender development initiatives, and their expertise in IT

systems. By including perspectives from different stakeholder groups, the study aims to develop a system that is not only functional but also aligned with the needs of its users. Table 1 below shows the distribution of the respondents.

Table 1. **Distribution of the Respondents**

Respondent	Frequency	Percentage
GAD Director	1	1.52%
GAD Secretary	1	1.52%
GAD Focal Persons	33	50%
RET Chairperson	1	1.52%
IT Experts	30	45.45%
Total	66	100%

The table above presents the distribution of the respondents involved in the study. Out of a total of 66 participants, the majority were GAD Focal Persons, comprising 50% of the sample. Their strong representation

highlights their central role in managing gender-related data and activities within their respective departments. IT Experts accounted for 45.45% of respondents, underscoring their vital function in evaluating the technical aspects and feasibility of the proposed GAD Record Management System. The GAD Director, GAD Secretary, and RET Chairperson were represented by one respondent each, making up 1.52% respectively. While fewer in number, their roles are pivotal in strategic decision-making and ensuring that the system aligns with the overall objectives of the GAD office.

Research Instrument

The research instruments used in this study are carefully designed to ensure the systematic collection of relevant data. These tools will assess the current manual system, guide the development process using the Modified Waterfall model of the Software Development Life Cycle (SDLC), and evaluate the quality and usability of the proposed web-based NEUST GAD Portal. The instruments will undergo reliability and validity testing prior

to full implementation to ensure the accuracy and consistency of the findings.

For statement number 1, to assess the existing manual record management system used by the NEUST GAD office, the study will employ survey questionnaires and semi-structured interviews with GAD focal persons and faculty members. The survey will focus on the following key areas:

Security of files and records measures how well sensitive data is stored and protected.

Management of data evaluates how records are collected, updated, and retrieved.

Reliability of output assesses the accuracy and consistency of generated reports.

Efficiency of processes determines how quickly and effectively data is processed and accessed.

Compliance with regulations and standards examines whether the current system meets institutional and legal requirements.

For SOP2, the Software Development Life Cycle (SDLC) specifically the Modified Waterfall model will be used to develop the web-based record management system. Each phase will involve specific data collection methods:

a. Requirements Gathering and Analysis Phase

Consultations with GAD officers and faculty members will be conducted to identify system requirements, define the project scope, and understand current challenges. Data will be collected through interviews, surveys, and document reviews to analyze user needs and inform system planning.

b. Design Phase

Based on the analyzed requirements, the system design will be developed using prototyping tools. UI mockups and system architecture diagrams will be created. User feedback will be gathered during this phase to ensure the design meets user expectations before development begins.

c. Implementation Phase

The system will be developed in line with

the approved design specifications. Developers may make limited revisions to the design if needed, based on technical feasibility or early feedback. Core functionalities will be prioritized to deliver a working version of the NEUST GAD Portal.

d. Testing Phase

The system will be rigorously tested using bug tracking tools, pilot user evaluations, and software quality checklists. Both functional and non-functional aspects will be validated to ensure the system's stability, performance, and user satisfaction. Necessary corrections will be implemented before deployment.

e. Deployment Phase

Following successful testing, the system will be officially launched for use across selected NEUST campuses. Orientation or brief training sessions may be held to support end-users. Deployment will be closely monitored to ensure a smooth transition and to address any immediate concerns.

f. Maintenance Phase

After deployment, the system will enter the maintenance phase, where updates, minor bug fixes, and enhancements will be performed based on ongoing feedback. Performance will be monitored continuously to keep the system aligned with user needs and institutional goals.

For SOP number 3, the system will undergo a software quality evaluation by IT experts, using a structured evaluation tool based on ISO 25002 software characteristics standards. The assessment will focus on:

1. Functional suitability focuses on how well the system meets the intended functions.
2. Performance efficiency assesses the speed and responsiveness of the system.
3. Compatibility measures the system's ability to integrate with existing platforms.
4. Usability checks the ease of navigation and accessibility of the system.

5. Reliability refers to the system's stability and ability to perform without failure.
6. Security focuses on the protection of sensitive data from unauthorized access.
7. Maintainability refers to the ease with which the software can be modified to correct faults, improve performance, or adapt to a changed environment.
8. Portability assesses the system's ability to be transferred from one hardware, software, or operational environment to another.

For SOP number 4, the final evaluation will focus on how end-users (GAD focal persons and faculty members) perceive the usability and effectiveness of the proposed system. A survey questionnaire will be distributed, assessing:

1. which the system provides functions that meet stated and implied user needs when used under specified conditions.

2. Performance Functional Suitability measures the degree to which the system's efficiency relates to the system's behavior regarding the processing time, responsiveness, and resource utilization under stated conditions.
3. Usability assesses the extent to which the system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.
4. Reliability refers to the system's ability to maintain a specified level of service, including maturity, availability, fault tolerance, and recoverability, under stated conditions over a specified period of time.
5. Security evaluates the degree to which the system protects information and data so that persons, other products, or systems have the data access appropriate to their types and levels of authorization.

The data collected from these research instruments will guide the refinement of the system, ensuring that it meets the needs of the NEUST GAD office and enhances gender development initiatives.

Data Gathering Procedures

Phase 1: Requirements Gathering and Analysis

The process begins with collecting detailed requirements from stakeholders, including GAD Director, Assistant/Technical Assistant, GAD focal persons, RET Chairpersons, and IT experts. Through structured surveys, interviews, and evaluations, both qualitative and quantitative data will be gathered to understand existing challenges in gender and development record management. Ethical approval will be secured from the NEUST administration and GAD office. Research instruments will be created and pilot-tested to ensure clarity and validity before full-scale data collection.

Phase 2: System Design

Using the analyzed requirements, the team will develop the system's architecture, interface layouts, database schema, and

process diagrams. A prototype will be designed to visualize the features and structure of the NEUST GAD Portal. This prototype will also be reviewed by selected users to gather early feedback before development begins.

Phase 3: Implementation

The actual development of the NEUST GAD Portal will take place in this phase. Developers will build the system functionalities as specified in the design documents. Regular collaboration with stakeholders will ensure the system aligns with user expectations. Updates and adjustments will be made iteratively as issues arise during development.

Phase 4: Testing

The completed system will undergo comprehensive testing to ensure reliability, performance, and user satisfaction. IT experts will evaluate technical aspects such as system efficiency, security, and maintainability, while User Acceptance Testing (UAT) will be conducted with GAD officers and faculty members to assess usability and practical effectiveness. Feedback will inform any necessary refinements.

Phase 5: Deployment

After successful testing and final adjustments, the NEUST GAD Portal will be deployed across selected campuses. Training sessions may be conducted to help users become familiar with the new system. The deployment will be closely monitored to ensure smooth transition from the manual system to the digital platform.

Phase 6: Maintenance

Post-deployment, the system will be regularly maintained to address any technical issues, implement updates, and enhance functionalities based on user feedback. Ongoing support will ensure the NEUST GAD Portal remains responsive to evolving needs and continues to support data-driven decision-making effectively.

Data Analysis

This study involves systematically examining the collected data to draw meaningful insights that address the research questions. The analysis aims to evaluate the

current manual system used by the NEUST GAD in managing gender development records and employee data. Using appropriate statistical tools and methods, the study will interpret the data to identify issues and challenges within the existing system. The results will help guide the development of a more efficient and reliable web-based solution, with a focus on enhancing data security, management, process efficiency, and compliance.

1. In analyzing the collected data, the researchers will employ frequency distribution and weighted means to interpret responses and evaluate the current manual system in terms of security of files and records, data management, output reliability, process efficiency, and compliance with regulations and standards. These statistical methods will summarize the feedback provided by faculty members and GAD focal persons, ensuring a clear representation of their perceptions and experiences with the existing manual system.

The results will be interpreted into Table 2 using the following scoring rubric, where responses will be categorized into

qualitative descriptions to ensure accurate assessment and meaningful insights.

Table 2. Scoring Rubrics for System Evaluation

Range	Qualitative Remarks	Verbal Description
3.26 - 4.00	Strongly Agree	The respondent has a high level of agreement with the statement or question. This means they strongly support or approve of the presented idea, concept, or statement.
2.51 - 3.25	Agree	This range signifies a moderate level of agreement. Respondents generally

Range	Qualitative Remarks	Verbal Description
		support the statement, but their approval is not as strong as those in the "Strongly Agree" category. This suggests that while they recognize the issue, there may be minor concerns or variations in their experiences.
1.76 - 2.50	Disagree	Scores in this range show that respondents do

Range	Qualitative Remarks	Verbal Description
1.00 - 1.75	Strongly Disagree	not support the statement. They express concerns or dissatisfaction with the described factor, indicating that the issue stated is a problem or needs improvement. This range represents a firm opposition to the statement. Respondents strongly reject the idea, demonstrating

Range	Qualitative Remarks	Verbal Description
		significant concerns or issues regarding the described aspect of the manual system.

The table 2 above presents the scoring rubric used to interpret the responses from the survey. It provides a quantitative assessment of the faculty members', GAD focal persons', and IT experts' perceptions of the current manual system in terms of security, data management, reliability, efficiency, and compliance with regulations.

2. For the second statement of the problem, the researchers will describe the activities for each phase of the Modified Waterfall Software Development Life Cycle model to provide relevant insights into the steps that will be undertaken for developing the NEUST GAD Portal. The Modified Waterfall model focuses on a structured approach to

system development that ensures the system's functionality is aligned with user needs. The phases of SDLC that will be addressed are as follows:

1. Requirements Gathering and Analysis Phase

In this phase, the researchers will define the project's scope, objectives, and limitations by engaging with stakeholders such as GAD focal persons and faculty members. Through surveys, interviews, and document reviews, the team will assess the current manual processes to identify inefficiencies and user needs. The collected data will help outline the functional and non-functional requirements of the proposed NEUST GAD Portal. These requirements will be validated with stakeholders to ensure they align with expectations and institutional goals.

2. Design Phase

Using the insights gained from the analysis phase, the researchers will develop detailed design documentation, including system architecture, database structure, and user interface mockups. Tools will be used to present interactive prototypes for initial feedback. This design will guide the technical specifications and define how the system

should behave to meet user expectations efficiently and effectively.

3. Implementation (Development) Phase

The development of the NEUST GAD Portal will begin based on the approved design specifications. Researchers will select appropriate programming languages and frameworks to code the system's core functionalities. Best practices in coding, modularity, and maintainability will be followed, with regular code reviews and limited iterations back to the design phase when necessary to accommodate practical coding adjustments.

4. Testing Phase

This phase involves a thorough evaluation of the system's performance, accuracy, and reliability. Multiple testing strategies will be used, such as unit testing for individual modules, integration testing for combined components, and user acceptance testing with GAD focal persons and faculty members. Feedback from these tests will be used to resolve bugs and fine-tune system behavior before the official launch.

5. Deployment Phase

After successful testing, the NEUST GAD Portal will be deployed to the live environment and made accessible to intended users across NEUST campuses. The deployment will include necessary orientation or training for users to ensure smooth adoption. This phase will be closely monitored to immediately address any unexpected issues.

6. Maintenance Phase

Following deployment, the system will enter ongoing maintenance to support its stability and adaptability. This includes monitoring performance, applying security patches, fixing bugs, and making system enhancements based on user feedback. Regular updates will ensure the system continues to meet evolving needs and remains aligned with institutional requirements.

Each of these phases will be crucial to the development and success of the NEUST GAD Portal. The researchers will ensure that the SDLC phases are executed methodically, guiding the system through its entire development cycle while addressing the challenges of data

security, process efficiency, and user satisfaction.

3. To address the third statement of the problem regarding the evaluation of the system's technical aspects, specifically the software's quality as per ISO 25010 standards, the researchers will utilize the evaluation criteria outlined in Table 3 for assessing Functional Suitability.

Table 3 presents the scoring rubric for Functional Suitability. Based on the evaluation outcomes from the IT experts, the following key software characteristics will be considered.

Table 3.

Scoring Rubric for Functional Suitability

Range	Quality Remarks	Verbal Description
3.26 - 4.00	Highly Suitable	The system meets or exceeds all functional requirements

Range	Quality Remarks	Verbal Description
		with high accuracy, completeness, and efficiency.
2.51 - 3.25	Suitable	The system meets most of the functional requirements with satisfactory accuracy, completeness, and efficiency.
1.76 - 2.50	Unsuitable	The system meets some of the functional requirements but with low accuracy,

Range	Quality Remarks	Verbal Description
		completeness , and efficiency.
1.00 - 1.75	Highly Unsuitable	The system fails to meet most or all of the functional requirements and has a deficient accuracy, completeness , and efficiency.

This table represents the scoring rubric used to evaluate the **Functional Suitability** of the system according to the IT expert's assessment.

Table 4 presents the scoring rubric for Performance Efficiency. Based on the evaluation outcomes from the IT experts, the following key software characteristics will be considered.

Table 4.

Scoring Rubric for Performance Efficiency

Range	Quality Remarks	Verbal Description
3.26 - 4.00	Excellent	The system performs its functions efficiently, with optimal use of resources, quick response times, and effective resource allocation.

Range	Quality Remarks	Verbal Description
2.51 - 3.25	Good	The system performs well with reasonable response times and efficient use of resources, but may have some minor inefficiencies.
1.76 - 2.50	Fair	The system shows moderate inefficiencies in response times, resource use, or capacity under certain conditions.

Range	Quality Remarks	Verbal Description
1.00 - 1.75	Poor	The system performs poorly, with slow response times, excessive use of resources, or an inability to meet system requirements.

This table presents the Performance Efficiency scoring rubric, used to assess how well the system utilizes resources under specific conditions, including factors like speed, resource allocation, and capacity.

Table 5 presents the scoring rubric for Compatibility. Based on the evaluation outcomes from the IT experts, the following key software characteristics will be considered.

Table 5.

Scoring Rubric for Compatibility

Range	Quality Remarks	Verbal Description
3.26 - 4.00	Highly Compatible	The system performs its operations efficiently while sharing resources and information seamlessly with other products or systems.
2.51 - 3.25	Compatible	The system works well with other products or systems but may have some minor integration

Range	Quality Remarks	Verbal Description
		issues or limitations.
1.76 - 2.50	Somewhat Compatible	The system has some compatibility issues that hinder its ability to work effectively with other systems.
1.00 - 1.75	Not Compatible	The system cannot interact effectively with other systems or software, preventing successful integration and resource

Range	Quality Remarks	Verbal Description
		sharing.

This table evaluates compatibility, which measures how well the system can exchange information and work with other software or hardware systems, ensuring that it can operate effectively in a shared environment.

Table 6 presents the scoring rubric for Usability. Based on the evaluation outcomes from the IT experts, the following key software characteristics will be considered.

Table 6.

Scoring Rubric for Usability

Range	Quality Remarks	Verbal Description
3.26 - 4.00	Excellent	The system is easy to learn, use,

Range	Quality Remarks	Verbal Description
		and navigate with a pleasant user interface. The system provides clear instructions and minimizes user errors.
2.51 - 3.25	Good	The system is relatively easy to use, but may have some minor usability issues, such as occasional difficulty in navigation or

Range	Quality Remarks	Verbal Description
1.76 - 2.50	Fair	slight interface improvements needed. The system requires considerable effort to learn or operate, and the interface is not very user-friendly, leading to increased errors or inefficiencies.

Range	Quality Remarks	Verbal Description
1.00 - 1.75	Poor	The system is difficult to learn and use, with a poorly designed interface and frequent user errors. The system does not meet basic usability standards.

This table evaluates Usability, which measures how well the system can be used to achieve specific goals effectively, efficiently, and satisfactorily. It focuses on user-friendliness, ease of learning, interface design, error prevention, and overall user satisfaction.

Table 7 presents the scoring rubric for Reliability. Based on the evaluation outcomes from the IT experts, the following key software characteristics will be considered.

Table 7.

Scoring Rubric for Reliability

Range	Quality Remarks	Verbal Description
3.26 - 4.00	Highly Reliable	The system performs its functions accurately and consistently under normal conditions. It has high availability, fault tolerance, and quick recoverability.
2.51 - 3.25	Moderately	The system

Range	Quality Remarks	Verbal Description
1.76 - 2.50	Reliable	mostly performs well, but may occasionally face minor issues with availability or fault tolerance, leading to minor downtime or interruptions.
	Slightly Reliable	The system experiences frequent downtimes or performance inconsistencies, causing disruption in operations. Its fault

Range	Quality Remarks	Verbal Description
		tolerance and recovery capabilities are limited.
1.00 - 1.75	Not Reliable	The system fails to perform reliably, with frequent errors, downtimes, and recovery issues. It is highly prone to system malfunctions and is unsuitable for use.

This table evaluates reliability, which measures the system's consistency in performing its functions under specified conditions for a defined period. It also focuses on the system's availability, fault tolerance, and ability to recover from failures.

Table 8 presents the scoring rubric for Security. Based on the evaluation outcomes from the IT experts, the following key software characteristics will be considered.

Table 8.

Scoring Rubric for Security

Range	Quality Remarks	Verbal Description
3.26 - 4.00	Strong Security	The system ensures high-level security with comprehensive protection against

Range	Quality Remarks	Verbal Description
2.51 - 3.25	Adequate Security	unauthorized access and data modification . It offers strong encryption, secure access, and maintains high integrity of data. The system provides satisfactory security measures, ensuring basic protection against unauthorized access.

Range	Quality Remarks	Verbal Description
1.76 - 2.50	Moderate Security	However, it may have minor vulnerabilities or areas that could be improved. The system has limited security features, which leave the system vulnerable to unauthorized access or data breaches. Protection mechanisms need substantial improvement.

Range	Quality Remarks	Verbal Description
1.00 - 1.75	Weak Security	The system provides insufficient security measures, making it highly vulnerable to unauthorized access, data corruption, and other security risks. Immediate improvements are necessary.

This table assesses Security, focusing on the system's ability to prevent unauthorized access, maintain data integrity, ensure confidentiality, and safeguard against potential threats or breaches.

Table 9 presents the scoring rubric for Maintainability. Based on the evaluation outcomes from the IT experts, the following key software characteristics will be considered.

Table 9.

Scoring Rubric for Maintainability

Range	Quality Remarks	Verbal Description
3.26 - 4.00	Excellent Maintainability	The system is highly maintainable, with well-documented code, clear modularity, and easy-to-troubleshoot components. Modifications and updates can be done with minimal

Range	Quality Remarks	Verbal Description
		impact on existing functionality .
2.51 - 3.25	Good Maintainability	The system is maintainable with some effort. Code is reasonably organized and maintainable, though there may be some complexity or dependencies that require additional work for updates or troubleshooting.
1.76 - 2.50	Fair Maintainability	The system has moderate

Range	Quality Remarks	Verbal Description
	ty	maintainability, requiring significant effort to update, troubleshoot, or modify. The code may be poorly structured or lacking documentation, which makes maintenance more challenging.
1.00 - 1.75	Poor Maintainability	The system is difficult to maintain and modify. The code is not well-structured, lacks

Range	Quality Remarks	Verbal Description
		documentation , and changes often introduce bugs or disrupt other parts of the system. Maintenance is time- consuming and costly.

This table evaluates the Maintainability of the system, focusing on the ease of modifying, updating, troubleshooting, and enhancing the system without disrupting existing features. It considers code structure, documentation quality, and how well the system can adapt to future requirements or issues.

Table 10 presents the scoring rubric for Portability. Based on the evaluation outcomes from the IT experts, the following key software characteristics will be considered.

Table 10.

Scoring Rubric for Portability

Range	Quality Remarks	Verbal Description
3.26 - 4.00	Highly Portable	The system can be easily moved or transferred to other environments or platforms without any glitches. It is highly adaptable to different hardware and software configurations.
2.51 - 3.25	Portable	The system can be moved to other environments or platforms with minimal effort. Some adjustments or configurations may be required for it to function optimally

Range	Quality Remarks	Verbal Description
1.76 - 2.50	Slightly Portable	in new environments. The system can be transferred to other environments or platforms, but it may require substantial modifications or troubleshooting. Some parts of the system may not function properly in different settings.
1.00 - 1.75	Not portable	The system is difficult to move or transfer to other environments or platforms. It requires significant changes to operate in a different environment, and some features may be incompatible with new configurations.

This table evaluates the **Portability** of the system, focusing on how well the system

can be transferred, installed, and operated in different environments or platforms. It considers the system's adaptability to new hardware, software configurations, and platforms.

7. To address the fourth statement of the problem, which concerns the evaluation of the system's quality by end-users, the researchers will utilize the following software characteristics guidelines. These guidelines are designed to enhance the evaluation outcomes and provide valuable insights into how the system performs based on the end users' experience and satisfaction with its key qualities.

Data Analysis Technique

Presented below are the formulas that the researchers will use in treating the data statistically.

Percentage Calculation

The percentage is used to calculate the rate of the responses from the respondents. To compute the percentage, the following formula will be used:

$$P = \frac{F}{N} \times 100$$

where:

P = Percentage

f = Frequency (Number of respondents selecting a particular response)

N = Total number of respondents

Weighted Arithmetic Mean

The weighted arithmetic mean is a statistical method used to calculate the average of responses, assigning different weights to different values. This is used to

determine how strongly respondents agree or disagree with statements.

The formula for computing the weighted mean is:

$$\text{Weighted Mean} = \frac{\sum(w \cdot x)}{\sum w}$$

Where:

www = Weight assigned to each response

xxx = Value of each response

$\sum(w \times x)$ = The sum of the weighted responses

$\sum w$ = The total sum of weights

Range of Weighted Mean and Verbal Interpretation

The ranges of the weighted mean and its corresponding verbal interpretation are shown in Table 11 below. This will be used to categorize the responses based on the weighted mean.

Table 11.

Response Mode

Range Of Weighted Mean	Verbal Interpretation
3.26 - 4.00	Strongly Agree
2.51 - 3.25	Agree
1.76 - 2.50	Disagree
1.00 - 1.75	Strongly Disagree

The researchers will use the weighted mean to analyze the feedback from the respondents and categorize it according to the above ranges. This will provide insights into the system's performance in terms of its quality characteristics based on respondents' evaluations.

By applying these formulas, the researchers will effectively analyze the data collected from faculty members, GAD focal persons, IT experts, and end-users, and will be able to generate meaningful insights to guide the development of the system.

Ethical Consideration

To ensure the protection of respondents' rights and maintain the integrity of the research process, the researchers will adhere to a strict set of ethical guidelines throughout the study. The following principles will guide the conduct of the study:

Informed Consent

Respondents will be provided with a detailed information sheet outlining the study's objectives, methods, potential risks, and benefits. They will have the opportunity to ask questions and clarify any concerns before agreeing to participate. Consent will be obtained voluntarily and in writing to ensure that the participants fully understand their involvement.

Anonymity and Confidentiality

The privacy of all respondents will be strictly maintained. Their identities will not be disclosed in any form of publication or reporting. All data will be anonymized, and access to sensitive information will be limited to authorized personnel only. The researchers will ensure that the data is

securely stored and handled to protect confidentiality.

Voluntary Participation

Participation in the study will be completely voluntary. Respondents will be informed that they can withdraw from the study at any point without facing any negative consequences. No pressure will be applied to encourage participation, and they will have the freedom to decide if they want to participate or not.

Protection of Vulnerable Populations

The researchers will take extra care to ensure that vulnerable populations, such as children and individuals with disabilities, are not exploited or exposed to unnecessary risks during the research. Safeguards will be put in place to protect their well-being.

Researchers' Integrity

The researchers will maintain a high level of integrity throughout the study. They will ensure that the study is conducted ethically, and the results are reported honestly and objectively. The research

findings will be presented without distortion, and no data will be manipulated.

In conclusion, the researchers are committed to adhering to ethical considerations such as informed consent, anonymity and confidentiality, voluntary participation, protection of vulnerable populations, and maintaining the integrity of the research process throughout the study.

CHAPTER 3

RESULTS AND DISCUSSION

This chapter provides a detailed analysis of the collected data. It involves a careful analysis followed by a detailed interpretation of the results. From this process, the goal is to draw out valuable insights of the research questions.

1. Assessment of the Manual System

The following tables display the assessment results. They detail the common problems identified with the manual system. This data outlines the key issues encountered.

1.1. Security of files and Records

Table 12 presents the assessment results from candidate end-users concerning the security of files and records. The table summarizes their collective evaluation.

Table 12

Result of the Assessment on the Security of
Files and Records Using Manual System

Item Statements	Weighted Mean	Verbal Description
The manual system is vulnerable to theft or unauthorized access.	3.04	Agree
There is a lack of control over who has access to files and records	3.09	Agree
The manual system is vulnerable to theft or unauthorized access.	3.11	Agree
The manual system does not provide adequate backup or disaster recovery options.	3.26	Strongly Agree

Item Statements	Weighted Mean	Verbal Description
The manual system is prone to errors, including data entry mistakes and transcription errors.	3.11	Agree
The manual system is difficult to update and maintain, leading to outdated or incorrect information	3.14	Agree
The manual system does not provide a complete audit trail of who has accessed or modified records.	2.70	Agree
The manual system does not provide	2.75	Agree

Item Statements	Weighted Mean	Verbal Description
the necessary confidentiality and privacy protections for sensitive information.		
The manual system is time consuming and labor-intensive to use and manage.	2.94	Agree
GRAND MEAN		3.01
Verbal		Agree

As illustrated in Table 12, the respondents generally viewed the manual system as having problems regarding the security of files and records, as shown in the computed grand mean of 3.01. Specifically, the respondents expressed that the manual system does not provide adequate backup or disaster recovery options ($\mu = 3.26$, Verbal Description = Strongly Agree) and is prone to errors,

including data entry mistakes and transcription errors ($\mu = 3.11$, Verbal Description = Agree). On the other hand, respondents also indicated that the manual system does not provide the necessary confidentiality and privacy protections for sensitive information ($\mu = 2.75$, Verbal Description = Agree).

According to Slyngstad et al. **(2022)**, manual record-keeping requires significantly more time for documentation and reporting compared to electronic systems, and it also increases the likelihood of errors due to variations in accuracy and data quality. Furthermore, a study in the Turkish Journal of Computer and Mathematics Education (2021) reported that paper-based documentation systems are prone to human error, slow information retrieval, and difficulty in updating, which can result in outdated or incorrect information. These studies highlight that the security and reliability of records in manual systems are compromised, reinforcing the respondents' perception of the manual system as vulnerable and inefficient.

The result of the assessment implies that respondents perceived the manual system as prone to errors, hard to maintain, and labor-intensive. This motivated the researchers to undertake this study because in a manual process, the handling of records by multiple employees heightens the risk of confidential information exposure. Additionally, one of the major drawbacks of manual processing is its inefficiency, which makes it difficult to manage and demands considerable effort.

1.2. Management of Data

Table 13 presents the candidate end-users' evaluation of the manual data management system. It displays their assessment results regarding its efficiency and organization. The findings highlight their collective experience with the current process.

Table 13

Results of the Assessment on the Management of Data Using Manual System

Item	Weighted Mean	Verbal
Statements		Description
The manual system results in errors in data entry.	3.30	Strongly Agree
The manual system leads to inefficient data retrieval and storage.	3.28	Strongly Agree
The manual system takes up too much time and resources.	3.26	Strongly Agree
The manual system lacks the ability to perform complex data analysis.	3.10	Strongly Agree

Item	Weighted Mean	Verbal
Statements		Description
The manual system is prone to data loss or corruption.	3.05	Agree
The manual system lacks data security and confidentiality.	2.95	Agree
The manual system lacks the ability to integrate with other systems.	2.90	Agree
The manual system requires manual processing and handling of paperwork	2.85	Agree

Item		Weighted Mean	Verbal
Statements			Description
The manual system lacks real-time data updates and reporting		2.80	Agree
The manual system lacks the ability to scale and handle larger volumes of data.		2.71	Agree
GRAND MEAN		3.00	
Verbal		Agree	

Table 13 shows that respondents found manual systems disputable for data management, as reflected in the general computed mean of 3.00. Specifically, the respondents concluded that the manual system results in errors in data entry ($\mu = 3.30$, Verbal Description = Strongly Agree) and leads to inefficient data retrieval and storage ($\mu = 3.28$, Verbal Description = Strongly Agree). On the other

hand, respondents also expressed that the manual system lacks the ability to scale and handle larger volumes of data ($\mu = 2.71$, Verbal Description = Agree).

Institutions are confronting data management challenges in manual systems. The manual system requires manual processing and handling of paperwork ($\mu = 2.85$, Verbal Description = Agree) and is prone to data loss or corruption ($\mu = 3.05$, Verbal Description = Agree). The respondents also noted limitations in complex data analysis ($\mu = 3.10$, Verbal Description = Strongly Agree), data security and confidentiality ($\mu = 2.95$, Verbal Description = Agree), integration with other systems ($\mu = 2.90$, Verbal Description = Agree), and real-time data updates and reporting ($\mu = 2.80$, Verbal Description = Agree).

The result of the assessment implies that the respondents perceived the manual system as causing errors, inefficiencies, and being difficult to scale or integrate. This suggests that reliance on manual data management depends heavily on the diligence of the staff, exposing records to risks such as

misplacement, damage, or loss. These findings underscore the need for more efficient and secure data management systems in institutions to minimize human error and enhance operational efficiency.

1.3. Reliability of the Output

Table 14 presents the candidate end-users' assessment of the system's output reliability. It displays their evaluation of the results' accuracy and dependability. The findings reflect their confidence in the system's performance.

Table 14

Results of the Assessment on the Reliability of Output Using Manual System

Item		Weighted Mean	Verbal
Statements			Description
The	manual	3.30	Strongly Agree
system			
produces			
inaccurate			
outputs.			

Item	Weighted Mean	Verbal
Statements		Description
The manual system is prone to errors in data entry.	3.28	Strongly Agree
The manual system is time consuming and inefficient.	3.11	Agree
The manual system requires significant effort to maintain and update.	3.23	Agree
The manual system lacks consistency in output quality.	3.00	Agree

Item Statements	Weighted Mean	Verbal Description
The manual system is susceptible to data loss and corruption.	2.95	Agree
The manual system is difficult to use for complex data analysis.	2.73	Agree
The manual system is not scalable to accommodate increased workload	2.93	Agree
The manual system is highly dependent on individual users' skills	3.23	Agree

Item Statements	Weighted Mean	Verbal Description
and knowledge.		
The manual system lacks security measures to protect confidential data.	3.27	Strongly Agree
GRAND MEAN	3.10	
Verbal		Agree

Table 14 shows that respondents perceived significant issues with the reliability of the manual system's output, as indicated by the overall computed mean of 3.10. Specifically, the respondents strongly concluded that the manual system produces inaccurate outputs ($\mu = 3.30$, Verbal Description = Strongly Agree) and is prone to errors in data entry ($\mu = 3.28$, Verbal Description = Strongly Agree). On the

other hand, respondents also agreed that the manual system is difficult to use for complex data analysis ($\mu = 2.73$, Verbal Description = Agree) and is susceptible to data loss and corruption ($\mu = 2.95$, Verbal Description = Agree).

Institutions are confronting output reliability challenges with manual systems. The manual system lacks consistency in output quality ($\mu = 3.00$, Verbal Description = Agree) and is highly dependent on individual users' skills and knowledge ($\mu = 3.23$, Verbal Description = Agree). The respondents also noted limitations in the system's ability to scale for increased workloads ($\mu = 2.93$, Verbal Description = Agree), and identified significant concerns regarding the lack of security measures for protecting confidential data ($\mu = 3.27$, Verbal Description = Strongly Agree).

The result of the assessment implies that the respondents perceived the manual system as unreliable, producing inconsistent and error-prone outputs while being vulnerable to security risks and data loss. This suggests that the integrity and trustworthiness of

outputs from a manual system are compromised by its inherent inefficiencies and lack of safeguards. These findings echo recent empirical evidence: for example, Garza et al. (2023) found that manual data processing methods had significantly higher error rates compared to more automated methods, underscoring the inaccuracy of manual recording. Likewise, in a comparison study of manual versus automated data collection in health settings, the automated approach substantially reduced transcription errors and improved data reliability (Yin et al., 2021). Such evidence reinforces the need for automated systems to ensure accuracy, consistency, and security of institutional data and reporting.

1.4. Efficiency of Processes

Table 15 presents the candidate end-users' assessment of the process efficiency within the manual system. It displays their evaluation of the workflow's speed and effectiveness. The findings reflect their direct experience with the system's operational pace.

Table 15

Results of the Assessment on the Efficiency of
Processes Using Manual System

Item Statements	Weighted	Verbal
	Mean	Description
The manual system is time-consuming and slows down the overall process.	3.27	Strongly Agree
Manual data entry is prone to errors and inaccuracies.	3.22	Agree
The manual system is inefficient and requires excessive use of paper and other resources.	3.26	Strongly Agree

Item Statements	Weighted Mean	Verbal Description
Manual systems are less reliable than computerized systems.	3.10	Agree
Manual systems lack the automation features necessary for efficient data management.	3.25	Strongly Agree
Manual systems require more personnel and resources than computerized systems.	2.94	Agree
Manual systems lack the flexibility to adapt to changing business needs and processes.	3.24	Agree

Item Statements		Weighted	Verbal
		Mean	Description
Manual systems are more vulnerable to security breaches and data loss than computerized systems.		2.75	Agree
Manual systems lack the analytical capabilities necessary for effective data analysis and reporting.		3.03	Agree
GRAND MEAN			3.11
Verbal			Agree

Table 15 shows that respondents identified significant inefficiencies in the manual system's processes, as indicated by the overall computed mean of **3.11**. Specifically,

the respondents strongly concluded that the manual system is time-consuming and slows down the overall process ($\mu = 3.27$, Verbal Description = Strongly Agree) and is inefficient in its excessive use of paper and other resources ($\mu = 3.26$, Verbal Description = Strongly Agree). On the other hand, respondents also agreed that manual systems require more personnel and resources than computerized systems ($\mu = 2.94$, Verbal Description = Agree) and are more vulnerable to security breaches and data loss ($\mu = 2.75$, Verbal Description = Agree).

Institutions are confronting critical operational challenges with manual systems. The system lacks the automation features necessary for efficient data management ($\mu = 3.25$, Verbal Description = Strongly Agree) and the flexibility to adapt to changing business needs ($\mu = 3.24$, Verbal Description = Agree). The respondents further confirmed that manual systems are less reliable than computerized systems ($\mu = 3.10$, Verbal Description = Agree) and lack the analytical capabilities for effective reporting ($\mu = 3.03$, Verbal Description = Agree).

The result of the assessment implies that the respondents perceived the manual system as a significant bottleneck, hindering productivity through its slow, resource-heavy, and inflexible nature. This suggests that the organization's operational agility is compromised, making it difficult to adapt to market changes or leverage data for strategic decisions. These findings are echoed in recent empirical research: for example, Haas et al. (2021) demonstrated that manual documentation in nursing management is notably time-consuming and error-prone, impairing workflow efficiency. Moreover, a 2021 study on manual inventory systems found that reliance on manual methods leads to wasted labor and error-driven inefficiencies (The Financial Impact of Manual Inventory Record Errors, 2021). Taken together, these studies reinforce the need to transition to an automated system to streamline workflows, reduce operational costs, and enhance overall process efficiency and resilience.

1.5. Compliance with Regulation and Standards.

Table 16 presents the candidate end-users' evaluation of the system's adherence to regulations and standards. It displays their assessment of its compliance and control measures. The findings reflect their confidence in the system's regulatory standing.

Table 16

Results of the Assessment on Compliance with Regulation and Standards Using Manual System

Item	Weighted Mean	Verbal
Statements		Description
Compliance with regulations and standards is difficult to maintain with a manual system.	3.30	Strongly Agree

Item	Weighted Mean	Verbal
Statements		Description
It is challenging to ensure that all staff are aware of and following compliance regulations and standards.	3.28	Strongly Agree
There is a risk of non-compliance due to human error in a manual system.	3.11	Agree
Manual systems lack the level of control necessary to ensure compliance with complex regulations and standards.	3.22	Agree

Item Statements	Weighted Mean	Verbal Description
Maintaining compliance with regulations and standards is time-consuming in a manual system.	3.00	Agree
Compliance requirements change frequently, making it difficult to keep up to date in a manual system.	2.80	Agree

Item	Weighted Mean	Verbal
Statements		Description
It is challenging to demonstrate compliance with regulations and standards in a manual system.	2.73	Agree
Manual systems are more prone to mistakes, which can lead to compliance issues.	3.21	Agree

Item Statements	Weighted Mean	Verbal Description
The lack of automation in a manual system makes it difficult to enforce compliance regulations and standards.	3.23	Agree
The absence of audit trails in a manual system makes it challenging to demonstrate compliance with regulations and standards.	3.25	Strongly Agree
GRAND MEAN	3.11	
Verbal		Agree

Table 16 shows that respondents perceive significant compliance risks in the manual system, as indicated by the overall computed mean of **3.11**. Specifically, the respondents strongly concluded that compliance with regulations is difficult to maintain ($\mu = 3.30$, Verbal Description = Strongly Agree) and that it is challenging to ensure all staff follow these protocols ($\mu = 3.28$, Verbal Description = Strongly Agree). On the other hand, respondents also agreed that demonstrating compliance is challenging ($\mu = 2.73$, Verbal Description = Agree) and that frequent changes in requirements are difficult to manage ($\mu = 2.80$, Verbal Description = Agree).

Institutions face substantial regulatory challenges with manual systems. The absence of reliable audit trails ($\mu = 3.25$, Verbal Description = Strongly Agree) and the lack of automation for enforcement ($\mu = 3.23$, Verbal Description = Agree) were identified as critical weaknesses. The respondents further confirmed that manual systems lack the necessary control for complex regulations (μ

= 3.22, Verbal Description = Agree) and are inherently prone to mistakes that create compliance issues (μ = 3.21, Verbal Description = Agree).

The result of the assessment implies that the manual system is viewed as a high-risk environment for regulatory adherence, largely due to its reliance on error-prone human processes and its lack of transparent, automated controls. This suggests that the organization's legal and operational integrity is consistently exposed to the risk of non-compliance, which can lead to penalties and reputational damage. These findings are reinforced by recent research: for example, Pagayonan (2021) found that manual record-keeping at a Philippine college made compliance with document control standards difficult—manual retrieval, version control, and retention requirements were not always met reliably under the manual regime. Likewise, an Industrial Case Study (Usman, Felderer, Unterkalmsteiner, Klotins, & Mendez, 2021) in a large-scale software development setting reported that compliance with changing standards (such as GDPR) was compromised by manual or semi-manual practices, due to

limited ability to audit, track protocol adherence, and respond quickly when regulations change. Such evidence strengthens the case for a digital system with built-in compliance checks, automated audit trails, and robust control frameworks to mitigate regulatory risks effectively.

2. The Design and Development of the System based on the Phases of the Software Development Life Cycle

Researchers completed the system's creation by working through these specific developmental stages.

2.1. Requirements Elicitation

During the requirements elicitation phase, the researchers established the project plan. This involved mapping the system's development activities and timelines onto a Gantt chart, guided by SDLC. To ground the design in real-world needs, surveys were distributed to candidate end-users and IT experts to gather insights on their processes and expectations.

Gantt Chart

A Gantt chart presents project activities as horizontal bars on a timeline, making it easy to read each task's start and end dates, durations, and points of overlap at a glance. This visual layout helps teams schedule work, monitor progress, and detect timing conflicts or dependencies quickly—so project managers can spot delays and reallocate effort or resources as needed (Khan et al., 2023). Empirical and case-based project-management research likewise treats Gantt charts as a standard scheduling/visualization technique that supports timeline estimation and overall progress monitoring in both traditional and modern projects. (Khan et al., 2023).

Figure 6 illustrates the project's development timeline, detailing the sequence of activities undertaken by the researchers.

Figure 6

Gantt Chart

2.2. Design Phase

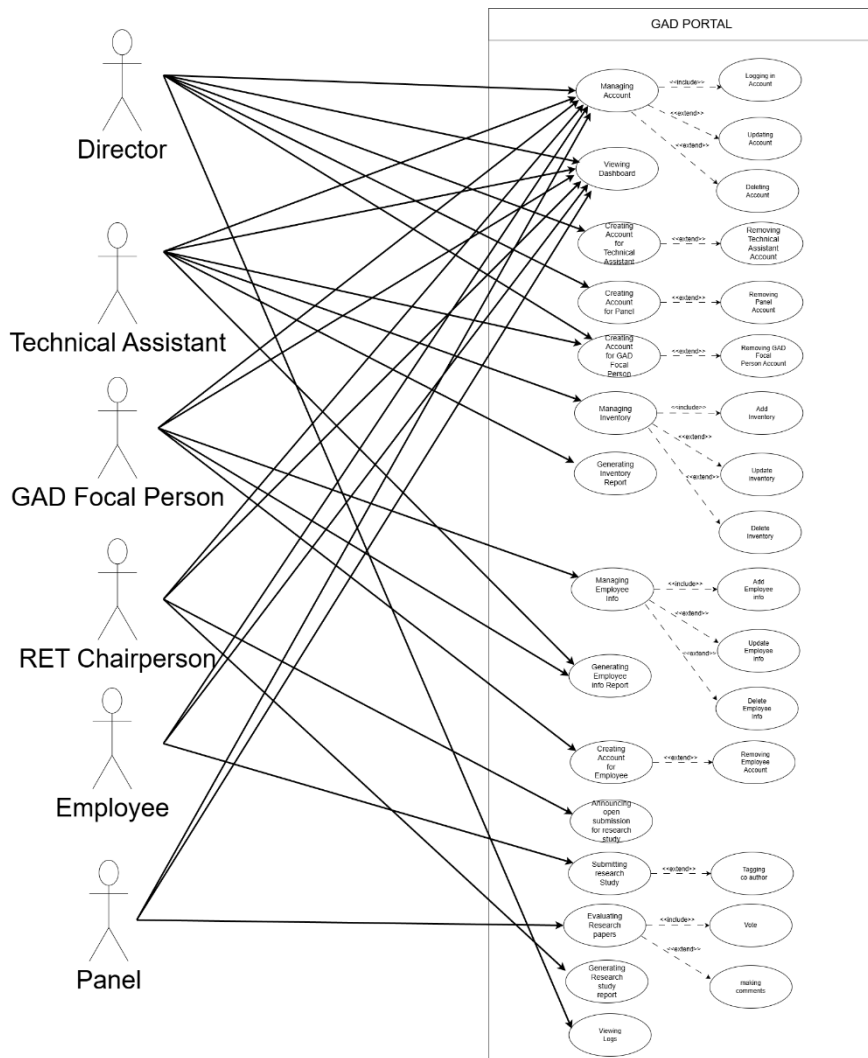
The design phase included creating the system's visual blueprint (wireframes and mockups) and its database design. Figure 7 shows the resulting use-case diagram, which displays all the system functions accessible to users.

Use-Case Diagram

Use-case diagrams serve as a visual blueprint, mapping the interactions between users (actors) and the specific functionalities a system offers.

Figure 7

Use-Case Diagram



The figure 7 illustrates how the actors interact with the GAD Portal System. The Director, as the super administrator, is responsible for managing user accounts, viewing dashboards, and monitoring system activities through the logs. The Director can also create and remove accounts for Technical Assistants, GAD Focal Persons, and Panels.

The Technical Assistant supports the Director by managing the inventory, generating inventory and employee reports, and creating accounts for GAD Focal Persons. The GAD Focal Person, on the other hand, manages employee records, updates employee information, and generates reports based on employee data. Additionally, the GAD Focal Person can create or elevate employee accounts within their respective departments.

Employees participate in the system by submitting research studies, viewing their research status, and engaging through comments. The RET Chair oversees research-related activities, including making announcements for research submissions and

updating grant statuses of research studies. The Panel evaluates submitted research and provides comments or feedback during the evaluation process.

Database Design

The process of database design organizes data within a chosen database model. By understanding the relationships between data, the designer determines what information needs to be stored. For this project, the researchers applied normalization techniques to identify dependencies between entities and their attribute

Database Normalization

Normalization is the method of structuring data within a database to ensure organization and efficiency. It involves designing tables and defining their relationships to reduce data redundancy and avoid inconsistent dependencies. The goal of this process is to safeguard data integrity and enhance database flexibility. By doing so, normalization ensures that information

is well-organized, making data retrieval and management more efficient and reliable

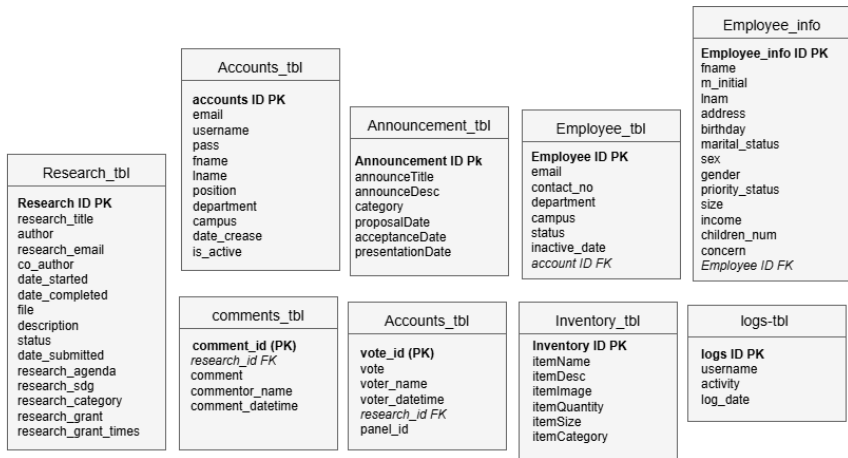
Figure 8

Database Normalization

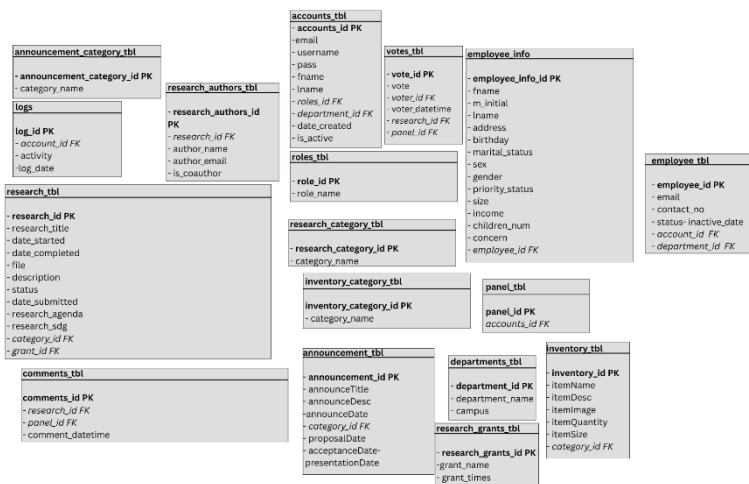
Unnormalized Form

account_id
email
username
pass
fname
lname
position
department
campus
date_created
is_active
announce_id
announceTitle
announceDesc
announceDate
category
proposalDate
acceptanceDate
presentationDate
employee_id
employee_email
contact_no
employee_department
employee_campus
status
inactive_date
employeeinfo_id
info_fname
m_initial
info_lname
address
birthday
marital_status
sex
gender
priority_status
size
income
children_num
concern
item_id
itemName
itemDesc
itemImage
itemQuantity
itemSize
itemCategory
log_id
log_username
activity
log_date
research_id
research_title
author
research_email
co_author
date_started
date_completed
file
description
research_status
date_submitted
research_agenda
research_sdg
research_category
research_grant
research_grant_times
comment_id
comment
commentor_name
comment_datetime
vote_id
vote
voter_name
voter_datetime
panel_id

Second Normal Form



Third Normal Form



The organizational context of the GAD Portal's database is shown in Figure 8. To

minimize data duplication, information pertaining to users is organized into separate, unique tables. This design principle ensures that each piece of data is maintained in a single location, leading to a more manageable and maintainable database. The effective implementation of this normalization process allows the GAD Portal to handle data efficiently, thereby ensuring the system's reliability in managing the research submission and approval lifecycle.

Entity-Relationship Diagram

An Entity-Relationship (ER) diagram models the conceptual structure of a database by illustrating the interconnections between its entities. The diagram comprises three core components: the entities themselves, their defining attributes, and the relationships that link them.

Figure 9

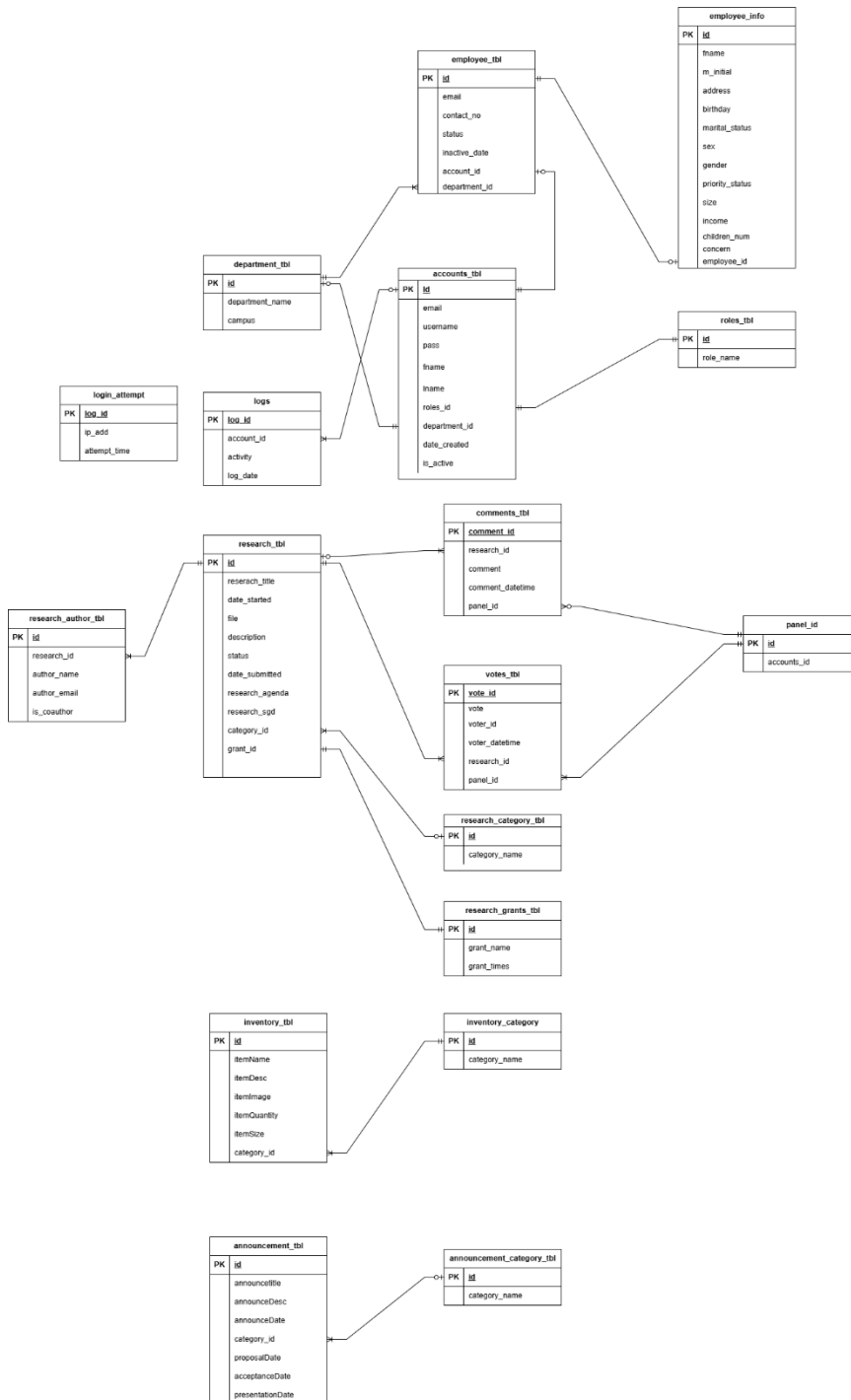


Figure 9 illustrates the entities and their relationships within the GAD Portal database. It presents how the different tables in the developed system are interconnected to support its core functionalities. The diagram comprises twenty-one (17) essential tables that manage various operations within the system. The accounts table contains information about user accounts, including roles such as Director, Technical Assistant, GAD Focal Person, Employee, RET Chair, and Panel. The employee table and employee table tables store detailed employee data, while the research table manages research submissions linked to related entities such as comments table, votes table, and research grants table. Additionally, inventory table and announcement table maintain records for inventory items and institutional announcements. Overall, this ERD provides a clear representation of the GAD Portal's database architecture, showing how the entities interact to support account management, research monitoring, and reporting processes.

Data Dictionary

The data dictionary for the GAD Portal, detailing the system's data structure, is presented in the tables below. Each entry includes the column name, data type, size, and a description of its function within the system.

Table 17

Data dictionary of accounts

Field Name	Data Type	Size	Description
account_id	INT	11	Primary key that uniquely identifies each user account and ensures data integrity

Field Name	Data Type	Size	Description
			across the system.
email	VARCHAR	255	User's email address used for system login, notifications, and official communications. Must be unique.
username	VARCHAR	50	Unique username for system authentication and user identification across all

Field Name	Data Type	Size	Description
			platform features.
pass	VARCHAR	255	Encrypted password stored securely using hashing algorithms for user authentication and data protection.
fname	VARCHAR	100	User's legal first name used for personalizing the interface

Field Name	Data Type	Size	Description
			and official documentation purposes.
lname	VARCHAR	100	User's legal last name used for complete identification in administrative records and reports.
roles_id	INT	11	Foreign key that defines user permissions and access levels by linking to

Field Name	Data Type	Size	Description
			the roles table.
department_id	INT	11	Foreign key that identifies the user's organizational unit for department-based filtering and reporting.
date_created	DATETIME	-	System-generated timestamp recording when the account was

Field Name	Data Type	Size	Description
			registered for audit and tracking purposes.
is_active	Boolean	-	Status indicator that controls account accessibility, where 1 enables login and 0 blocks system access.

Table 18

Data dictionary of employee info

Field Name	Data Type	Size	Description
employee_in fo_id	INT	11	Primary key that uniquely identifies each employee's personal information record in the system.
fname	VARCHAR	100	Employee's legal first name as it appears on official documents and identification cards.

Field Name	Data Type	Size	Description
m_initial	CHAR	1	Employee's middle initial used for proper name formatting and official documentation purposes.
lname	VARCHAR	100	Employee's legal last name or surname used for complete identification in all official records.

Field Name	Data Type	Size	Description
address	VARCHAR	255	Complete residential address of the employee for contact purposes and official correspondence.
birthday	DATE	-	Employee's date of birth used for age verification and demographic analysis in GAD reporting.

Field Name	Data Type	Size	Description
marital_status	VARCHAR	20	Current marital status of the employee for demographic profiling and benefits administration.
sex	VARCHAR	10	Biological sex of the employee used for gender-disaggregated data analysis and reporting.

Field Name	Data Type	Size	Description
gender	VARCHAR	20	Gender identity of the employee essential for comprehensive gender and development analysis.
priority_status	VARCHAR	50	Special classification status relevant to GAD programs and targeted interventions.

Field Name	Data Type	Size	Description
size	VARCHAR	20	Clothing or uniform size information for administrative and procurement requirements .
income	DECIMAL	10,2	Monthly income data used for socioeconomic analysis and GAD program assessment.

Field Name	Data Type	Size	Description
children_num	INT	2	Number of children the employee has for family-related benefits and program planning.
concern	TEXT	-	Specific notes or concerns related to the employee's GAD-related needs or circumstances.

Field Name	Data Type	Size	Description
employee_id	INT	11	Foreign key that links to the main employee table for employment record connection.

Table 19

Data dictionary of research

Field Name	Data Type	Size	Description
research_id	INT	11	Primary key that uniquely

Field Name	Data Type	Size	Description
			identifies each research study submitted to the GAD Portal system.
research_title	VARCHAR	255	Official title of the research study as submitted by the researcher for identification.

Field Name	Data Type	Size	Description
date_started	DATE	-	Start date of the research project indicating when the research work officially commenced.
date_completed	DATE	-	Completion date when the research study was finalized and ready for submission.
file	VARCHAR	255	File path or name of the

Field Name	Data Type	Size	Description
			uploaded research document in PDF or document format.
description	TEXT	-	Comprehensive abstract or summary describing the research objectives and methodology.
status	VARCHAR	50	Current status in the review workflow (Pending,

Field Name	Data Type	Size	Description
			Under Review, Approved, Rejected) .
date_submitted	DATE TIME	-	Date and time when the research was formally submitted to the GAD Portal system.
research_agenda	VARCHAR	100	Specific GAD research agenda category that the research study

Field Name	Data Type	Size	Description
			primarily addresses.
research_sd g	VARCHAR	100	Sustainable Development Goal(s) that the research study aligns with and supports.
category_id	INT	11	Foreign key linking to research category table for thematic classification of the study.

Field Name	Data Type	Size	Description
grant_id	INT	11	Foreign key referencing grants table if the research received GAD funding support.

Table 20

Data dictionary of inventory

Field Name	Data Type	Size	Description
inventory_id	INT	11	Primary key that uniquely

Field Name	Data Type	Size	Description
			identifies each inventory item in the GAD Portal system.
itemName	VARCHAR	100	Official name of the inventory item used for identification and tracking purposes.
itemDesc	TEXT	-	Detailed description of the inventory item

Field Name	Data Type	Size	Description
			including features and intended usage.
itemImage	VARCHAR	255	File path or URL for the item's visual representation in the inventory system.
itemQuantity	INT	11	Current stock quantity of the item for inventory management and

Field Name	Data Type	Size	Description
			monitoring purposes.
itemSize	VARCHAR	50	Physical dimensions or capacity specifications of the inventory item when applicable.
category_id	INT	11	Foreign key linking to category table for organizing items by type and purpose.

Table 21*Data dictionary of announcement*

Field Name	Data Type	Size	Description
announcement_id	INT	11	Primary key that uniquely identifies each announcement within the GAD Portal system.
announcementTitle	VARCHAR	200	Main headline or subject line of the announcement displayed to

Field Name	Data Type	Size	Description
			all system users.
announceDesc	TEXT	-	Complete content and detailed information contained within the announcement message.
announceDate	DATE TIME	-	Publication date and time when the announcement becomes visible to system users.

Field Name	Data Type	Size	Description
category_id	INT	11	Foreign key linking to category table for organizing announcements by type and audience.
proposalDate	DATE	-	Deadline or important date for research proposal submissions as specified in announcement.

Field Name	Data Type	Size	Description
acceptanceDate	DATE	-	Scheduled date for releasing research acceptance notifications to submitting authors.
presentationDate	DATE	-	Planned date for research presentations or project deliberations as announced to participants.

Table 22

Data dictionary of employee

Field Name	Data Type	Size	Description
employee_id	INT	11	Primary key that uniquely identifies each employee record within the GAD Portal system.
email	VARCHAR	255	Official work email address of the employee used for organizational

Field Name	Data Type	Size	Description
			communications.
contact_no	VARCHAR R	20	Contact number of the employee for official correspondence and emergency purposes.
status	VARCHAR R	20	Current employment status indicating whether the employee is active or inactive.

Field Name	Data Type	Size	Description
inactive_date	DATE	-	Date when the employee was deactivated or separated from the organization .
account_id	INT	11	Foreign key linking to the accounts table for system access and authentication purposes.
department_id	INT	11	Foreign key referencing the

Field Name	Data Type	Size	Description
			department table for organizational structure mapping.

Table 23

Data dictionary of research author

Field Name	Data Type	Size	Description
research_authors_id	INT	11	Primary key that uniquely identifies each author entry in the

Field Name	Data Type	Size	Description
			research authorship records.
research_id	INT	11	Foreign key linking to the research table to associate authors with specific studies.
author_name	VARCHAR	255	Full name of the research author or co-author as it should appear in

Field Name	Data Type	Size	Description
			publications .
author_email	VARCHAR	255	Email address of the author used for correspon- ce and research- related communicatio- ns.
is_coauthor	TINYINT	1	Flag indicating whether the author is a co-author (1) or primary

Field Name	Data Type	Size	Description
			author (0) of the research.

Table 24

Data dictionary of department

Field Name	Data Type	Size	Description
department_id	INT	11	Primary key that uniquely identifies each department within the organization

Field Name	Data Type	Size	Description
			's structure.
department_name	VARCHAR	100	Official name of the department or organizational unit for identification and reporting.
campus	VARCHAR	50	Campus or location where the department is primarily based for

Field Name	Data Type	Size	Description
			organizational mapping.

Table 25

Data dictionary of roles

Field Name	Data Type	Size	Description
role_id	INT	11	Primary key that uniquely identifies each user role within the GAD Portal system.

Field Name	Data Type	Size	Description
role_name	VARCHAR	50	Name of the user role defining permission levels and system access privileges.

Table 26

Data dictionary of panel

Field Name	Data Type	Size	Description
panel_id	INT	11	Primary key that uniquely

Field Name	Data Type	Size	Description
			identifies each panel member in the research evaluation committee.
accounts_id	INT	11	Foreign key linking to user accounts table to associate panel members with system access.

Table 27

Data dictionary of panel

Field Name	Data Type	Size	Description
comments_id	INT	11	Primary key that uniquely identifies each comment made on research submissions.
research_id	INT	11	Foreign key linking to the research table to associate comments with specific studies.

Field Name	Data Type	Size	Description
panel_id	INT	11	Foreign key identifying the panel member who submitted the comment or feedback.
comment_datetime	DATETIME	-	Date and time when the comment was submitted by the panel member for tracking.

Table 28

Data dictionary of inventory category

Field Name	Data Type	Size	Description
inventory_category_id	INT	11	Primary key that uniquely identifies each category in the inventory classification system.
category_name	VARCHAR	100	Name of the inventory category used for organizing and classifying inventory items.

Table 29

Data dictionary of announcement category

Field Name	Data Type	Size	Description
announcement_category_id	INT	11	Primary key that uniquely identifies each category for organizing systems.
category_name	VARCHAR	100	Name of the announcement category used to classify announcements by purpose

Field Name	Data Type	Size	Description
			and audience.

Table 30

Data dictionary of research category

Field Name	Data Type	Size	Description
research_category_id	INT	11	Primary key that uniquely identifies each research category in the

Field Name	Data Type	Size	Description
			classification system.
category_name	VARCHAR	100	Name of the research category used to classify studies by topic and research focus area.

Table 31

Data dictionary of research grants

Field Name	Data Type	Size	Description
research_grants_id	INT	11	Primary key that uniquely identifies each research grant available in the system.
grant_name	VARCHAR	150	Official name of the research grant or funding program for

Field Name	Data Type	Size	Description
			identificati on purposes.
grant_times	INT	11	Number of times the grant has been awarded or is available for allocation to research.

Table 32

Data dictionary of votes

Field Name	Data Type	Size	Description
vote_id	INT	11	Primary key that uniquely identifies each vote cast in the research evaluation process.
vote	VARCHAR	20	The actual vote value cast by the panelist (e.g., Approve,

Field Name	Data Type	Size	Description
			Reject, Revise) .
voter_id	INT	11	Foreign key identifying the panel member who cast the vote on the research submission.
voter_datetime	DATETIME	-	Date and time when the vote was submitted by the panel member for audit purposes.

Field Name	Data Type	Size	Description
research_id	INT	11	Foreign key linking to the research table to associate votes with specific studies.

Table 33

Data dictionary of logs

Field Name	Data Type	Size	Description
log_id	INT	11	Primary key that uniquely identifies each activity log entry in the system.
account_id	INT	11	Foreign key linking to the user account that performed the recorded activity.

Field Name	Data Type	Size	Description
activity	VARCHAR	255	Description of the user action or system event that was logged for auditing.
log_date	DATE TIME	-	Date and time when the activity was performed and recorded in the system.

The data dictionary is a centralized repository of metadata that describes the data structure of the GAD Portal. It serves as a critical reference for developers and

administrators, facilitating database management, ensuring data integrity, and providing a clear understanding of the system's underlying architecture.

2.3. Development

The development phase centers on implementing the system's functional and non-functional requirements. This process is visualized using a Data Flow Diagram (DFD), which models the flow of information through the system using its core components: external entities, processes, data flows, and data stores.

Context Diagram

A Context Diagram represents the highest level of a Data Flow Diagram (DFD), providing a conceptual overview of a system's data exchanges. It depicts the entire system as a single process, focusing on the data flows between it and external entities, while abstracting all internal processes.

Figure 10

Context Diagram

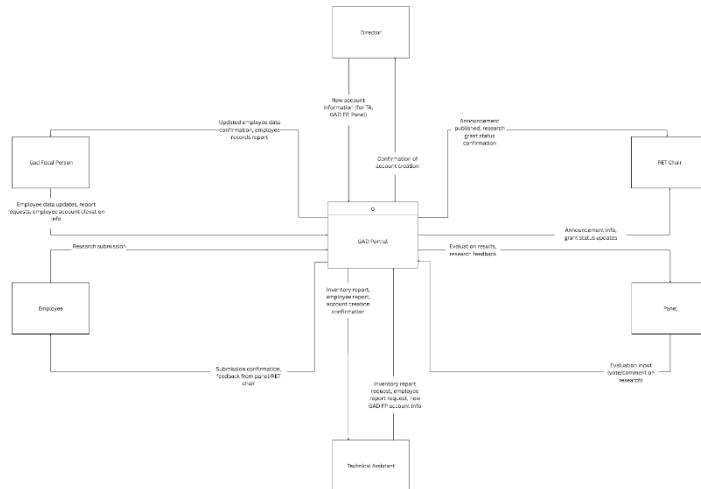


Figure 10 provides a high-level representation of the system's interaction. The main actors are the **Director**, **Technical Assistant**, GAD Focal Person, Employee, RET Chair, and Panel.

The **Director** engages with the system to create accounts for other key roles, and the system returns a confirmation of each account creation. The **Technical Assistant** and **GAD Focal Person** partake in administrative functions; the Technical Assistant inputs requests for inventory and employee reports and can create new GAD Focal Person accounts, receiving the corresponding reports and creation confirmations in return. The GAD Focal Person inputs employee data and can elevate this information into a formal system

account, with the system outputting confirmations and employee records.

On the research side, Employees interact with the system by submitting research, and in return, they receive submission confirmations and feedback. The RET Chair inputs announcements and grant status confirmations, and the system outputs this published information and status updates. Finally, the Panel engages by providing evaluations and comments on research papers, and the system outputs an evaluation confirmation and relays their feedback.

The diagram depicts the scope of the system, showing how it centralizes the management of employee accounts, administrative reports, and the entire research submission and evaluation workflow, highlighting the vital features within the study's context.

Data Flow Diagram - Level 0

A Level 0 DFD provides a high-level overview by identifying a system's key processes and showing the data flows between them, external entities, and data repositories

Figure 11.1

Data Flow Diagram - Level 0

Managing Accounts

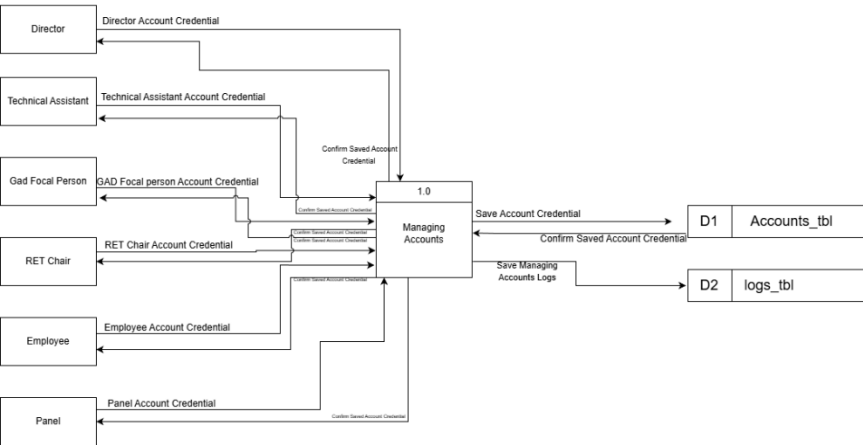


Figure 11.2

Viewing Dashboard

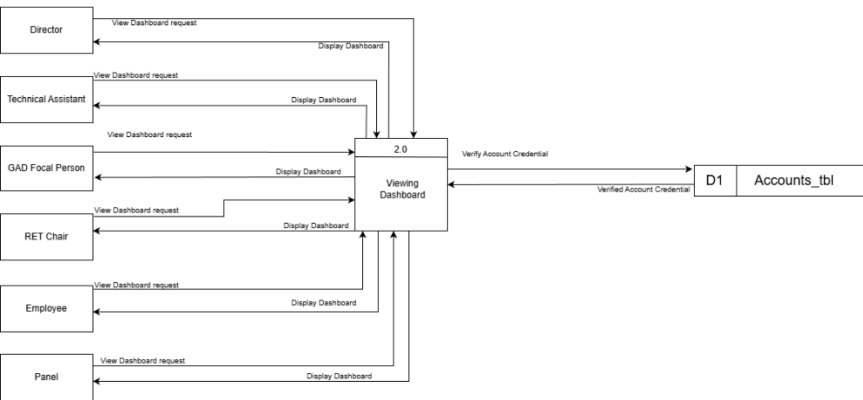


Figure 11.3

Account Creation

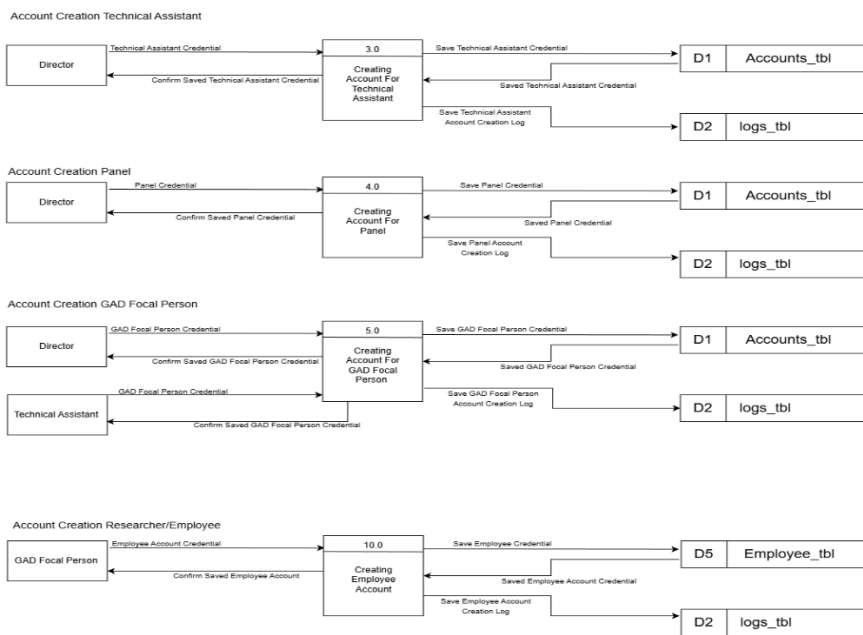


Figure 11.4

Inventory/employee record management

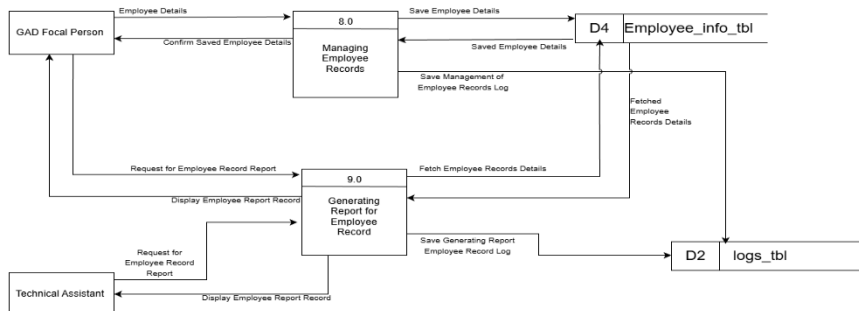
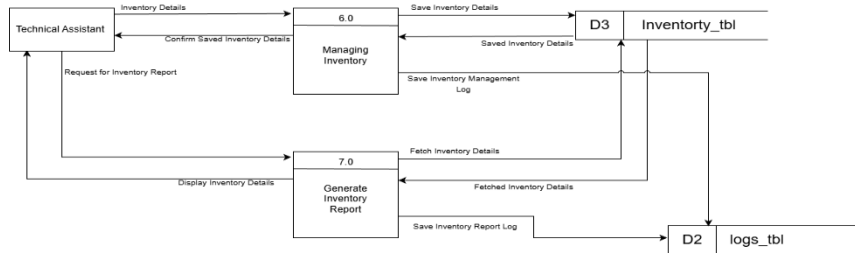


Figure 11.5

Research Submission

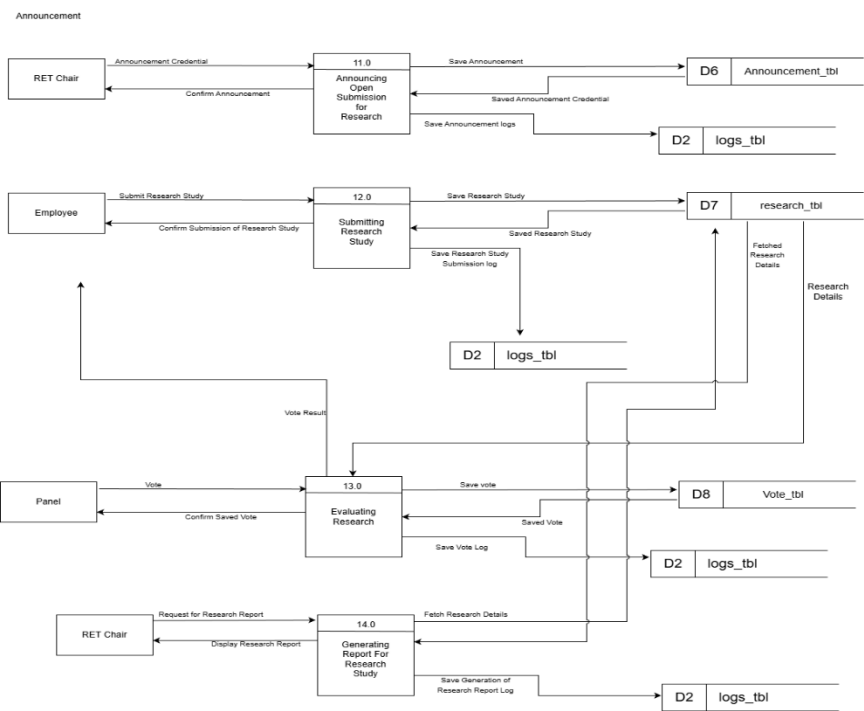


Figure 11.6

Viewing logs

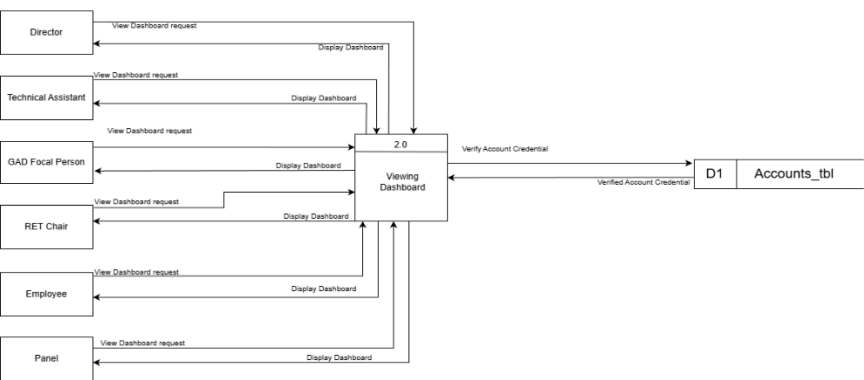


Figure 11 presents how data flows within the GAD Portal System. DFD Level 0 illustrates the main processes, including managing user accounts, creating other user like, technical assistant, GAD Focal Person, Panel, and Employee, handling inventory and employee information, managing research submissions, posting announcements, recording votes, and monitoring system logs. The diagram shows how data moves between users such as the Director, Technical Assistant, GAD Focal Person, RET Chair, Employees, and Panels, and the corresponding data stores. This provides an overview of how each module interacts and supports the overall operations of the GAD Portal.

Data Flow Diagram - Level 1

A Level 1 Data Flow Diagram elaborates on the system's structure by decomposing its primary function into constituent sub-processes, thereby illustrating the internal components with greater detail than the higher-level context diagram.

Figure 12.1

Data Flow Diagram - Level 1

Account management

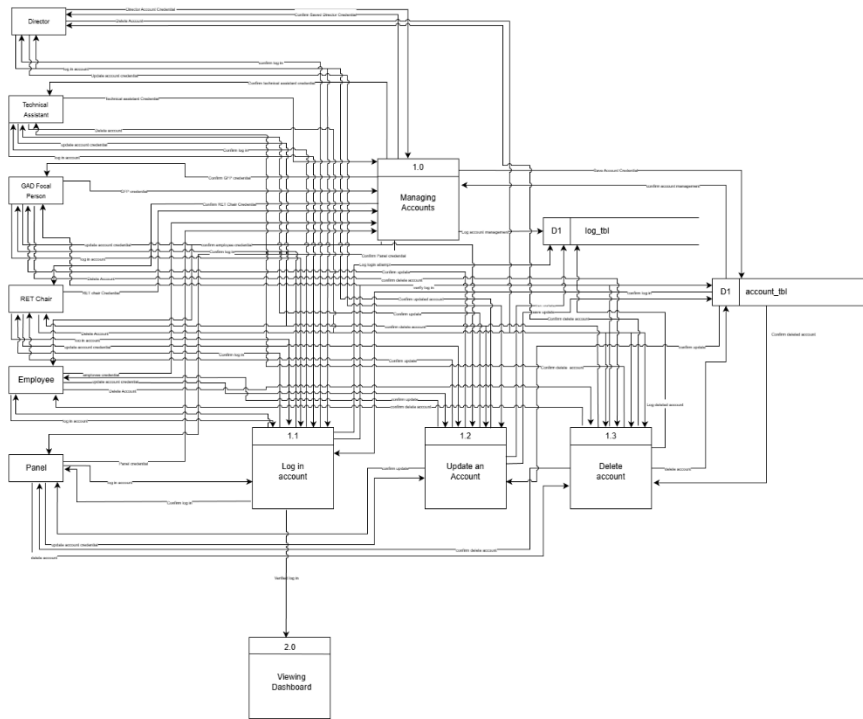
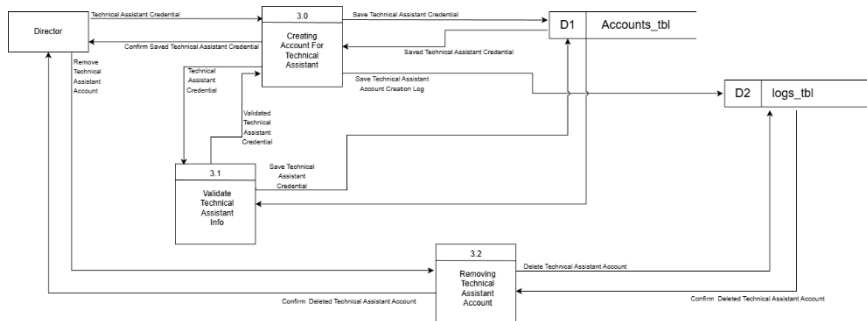


Figure 12.2

Account Creation



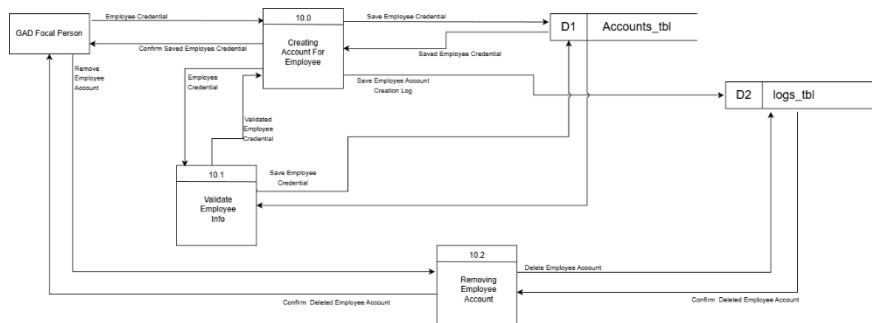
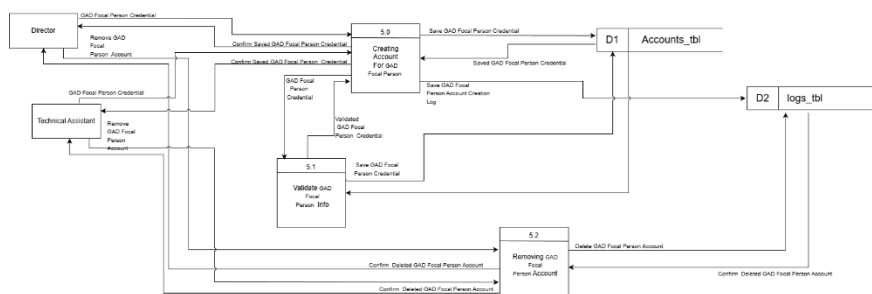
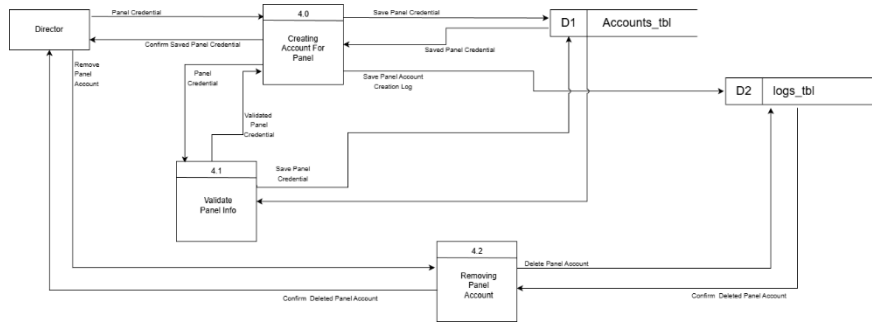


Figure 12.3

Inventory Management

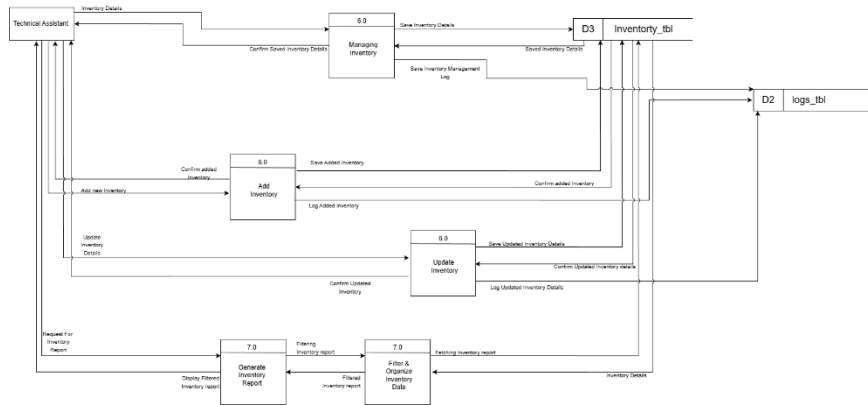


Figure 12.4

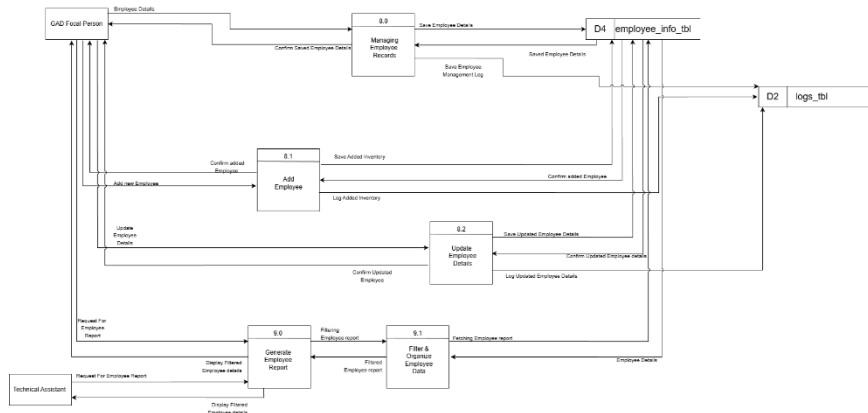


Figure 12.5

Research Submission

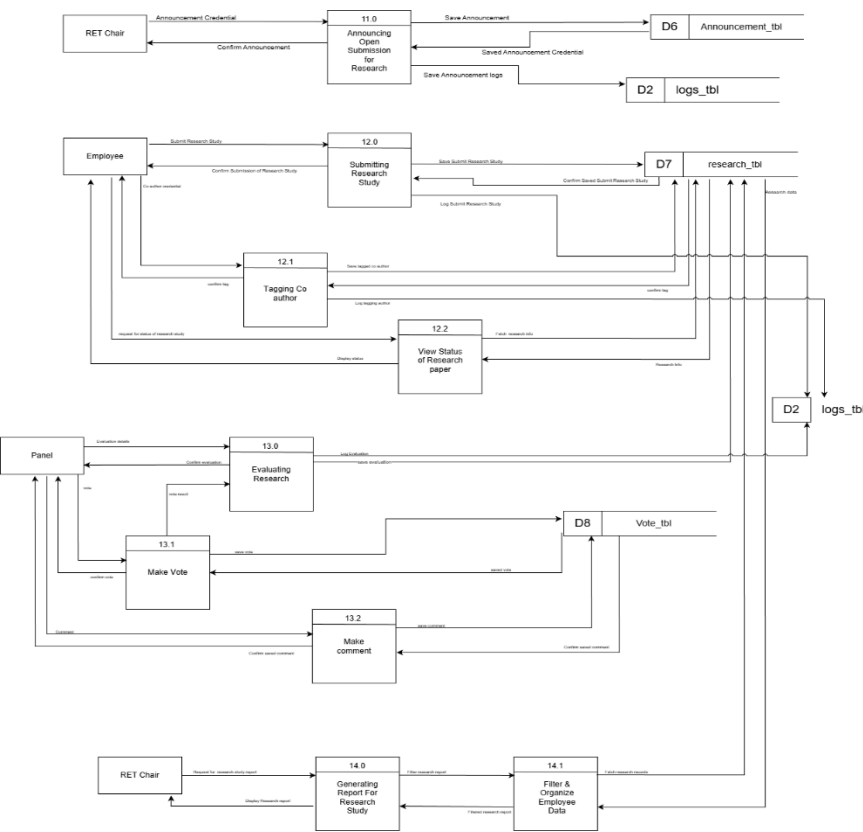


Figure 12.6

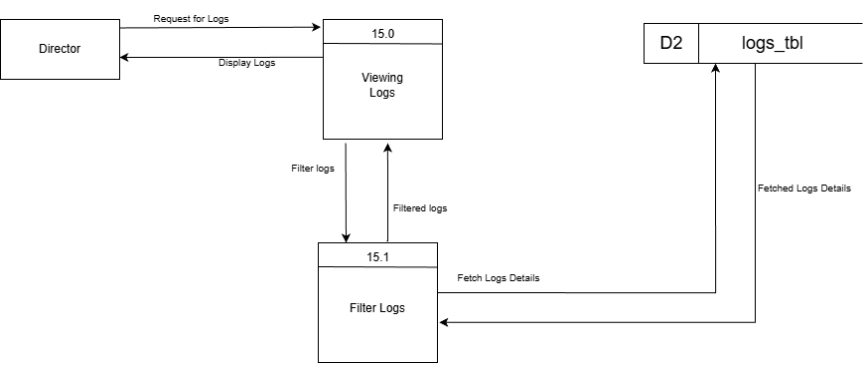


Figure 12 shows the data flow within the sub-processes of the GAD Portal system. The Managing Account process is divided into sub-processes such as logging into an account updating account information and deleting account. All user authentication attempts and successful transactions are recorded in the `logs` table, with account data being stored in the `accounts` table.

The View Dashboard process has no sub-process but serves as a central hub where users can access and view summarized data relevant to their roles.

The Creating Technical Assistant Account process, accessible only to the Director, is composed of validating the account information and removing a technical assistant account, with all changes saved in the `accounts` table. The Creating Panel Account process follows the same sub-processes and also stores data in the same table. Similarly, the Creating GAD Focal Person Account process, which can be done by both the Director and Technical Assistant, includes validating and removing GAD Focal

Person accounts, with corresponding records saved in the accounts table.

The Managing Inventory process, handled by the Technical Assistant, includes adding and updating inventory items, which are then stored in the inventory table. The Generating Inventory Report process involves filtering and organizing inventory data retrieved from the same table.

The Generating Employee Report process, which can be performed by both the Technical Assistant and the GAD Focal Person, involves filtering and organizing employee data obtained from the employee info table. The Managing Employee Records process includes adding and updating employee records, which are saved in the same table.

The Creating Employee Account process, performed by the GAD Focal Person, includes archiving or deleting employee accounts, with results reflected in the employee table. The Submitting Research Study process, carried out by employees, is composed of tagging co-authors and submitting the research, where research data are stored in the research table and co-

author data are linked to the `employee` table.

The Announcing Open Submission for Research process, handled by the RET Chair, has no sub-process but allows the publication of open calls for research submissions. The Evaluating Research process, done by the Panel, includes making comments and recording evaluations. The comments are stored in the `comments` table, while evaluations are stored in the `votes` table.

The Generating Research Report process, performed by the RET Chair, retrieves and organizes data from the `research` table. Finally, the Viewing Logs process, accessible only to the Director, includes filtering activity records that are fetched from the `logs` table. Every system action performed by any user, including login attempts, is recorded in this table to ensure comprehensive accountability and traceability.

2.4. Testing

The researchers evaluated the GAD Portal's compliance with both functional and nonfunctional requirements during this phase. Black-box testing was selected as the primary method for the participating IT Experts. This approach ensured that testers had no prior knowledge of the system's internal architecture or business logic, allowing them to identify potential issues from an external user's perspective before any internal inspection was conducted.

For the evaluation involving candidate end-users, acceptance testing a subset of functional testing was employed. The researchers collected feedback from these users to assess how the system performed in real-world scenarios. This input served as a valuable reference for refining and enhancing the GAD Portal based on practical user experience and operational context.

2.5. Deployment

Following the successful completion of the testing phase, the GAD Portal will progress

to deployment within a live operational environment. This critical stage encompasses two primary activities: first, the technical preparation of the production environment, including server configuration, database setup, and network integration to ensure stable performance; and second, securing final approval from the candidate end-users, confirming that the system meets their operational needs and expectations.

The primary cost will be associated with hosting the system on a web server. Estimated expenses are confined to securing a professional domain name (approximately ₱500-₱1,000 annually) and a reliable hosting service (ranging from ₱0 for a free-tier plan to around ₱5,000 annually for a robust shared hosting plan).

This deployment process is contingent upon the successful execution of all test cases. The comprehensive test suite was designed to rigorously validate the system's functionality—ensuring features like research submission, panel evaluation, and report generation work as intended—its accuracy in data processing and reporting,

and its full compliance with established software standards for security, usability, and reliability.

Once the system has passed this stringent verification and all stakeholders have provided their approval, the deployment phase will commence. This signifies that the GAD Portal has been thoroughly vetted and is officially ready for active use by employees, panelists, and administrators to manage the institution's Gender and Development research lifecycle.

2.6. Maintenance

During this phase, the researchers concentrate on correcting software defects and refining the system's architecture to guarantee proper functionality. The work involves implementing several types of software maintenance, with a primary focus on:

- 1. Corrective Maintenance** Addressing any bugs, errors, or vulnerabilities discovered after the GAD Portal goes live. This includes debugging submission workflows, fixing miscalculations in reports, and

resolving any system crashes to ensure stable daily operation.

2. **Perfective Maintenance** Enhancing the system based on user feedback to better meet the evolving needs of the GAD Focal Person, panelists, and employees. This involves adding new features, such as additional report filters, improving the user interface (UI) for the research submission form, or optimizing database queries to speed up loading times.
3. **Preventive Maintenance:** Proactively improving the codebase to ensure the system's long-term health and maintainability. This includes activities like refactoring complex code sections, updating internal documentation, and cleaning the database to prevent future performance issues or bugs.

REFERENCES

- Asana, T. (2024, July 3). Data-Driven Decision Making: A Step-by-Step Guide [2024] • Asana. Asana.
<https://asana.com/resources/data-driven-decision-making>
- B., D., & A., A. (2025). Paper based system of record keeping and organizational performance in Mitooma District in Western Uganda. *International Journal of Multidisciplinary Contemporary Education Research (IJMCER)*, *6*(1), 40-52
- Baek, J., Lawson, J., & Rahimzadeh, V. (2024).
Investigating the roles and responsibilities of institutional signing officials after data sharing policy reform for federally funded research in the United States: National Survey. JMIR Formative Research, 8, e49822. <https://doi.org/10.2196/49822>

Belas, J., Kliestik, T., Dvorsky, J., & Streimikiene, D. (2025).

Exploring gender-based disparities in the digital transformation and sustainable development of SMEs in V4 countries. *Journal of Innovation & Knowledge*, 10(2), 100681.

<https://doi.org/10.1016/j.jik.2025.100681>

Berezhna, G. V., Aleinikova, O. V., Kovtun, O. A., Danylchuk, H. B., Babenko, V. O., & Nechypurenko, P. P. (2025).

Educational technology for gender mainstreaming in project management: Advances in training approaches and digital tools. *CTE Workshop Proceedings*.

<https://doi.org/10.55056/cte.669>

Box, D., &. (2022, November 9). STUDY DOCUMENTS THE ADVANTAGES OF AN AUTOMATED RECORDS MANAGEMENT SYSTEM.
<https://www.linkedin.com/pulse/study->

[documents-advantages-automated-records-management-system-](#)

Data Management Security | What it is and why it's important. (2025, July 27). Cohesity.

<https://www.cohesity.com/glossary/data-management-security/>

ESCAP. (2023). *Progress report on gender mainstreaming 2023:*

Advancing gender equality in a post-pandemic world. United Nations Economic and Social Commission for Asia and the Pacific.

https://www.asiapacificgender.org/sites/default/files/2024-03/ESCAP%20Gender%20Mainstreaming%20Report%202023_FINAL-compressed.pdf?utm_source

European Union Delegation to the Philippines. (2021).

Gender country profile for the Philippines 2021 (Contract No. 2021/424-784).

https://niccdies.climate.gov.ph/files/documents/gender_country_profile_2021_philippines.pdf?utm_source

Claude, K. J. (2024). The impact of records management on the effectiveness of human resource management in the public sector. *International Journal of Research and Innovation in Social Science*, VIII(V), 1040-1060.
<https://doi.org/10.47772/ijriss.2024.805072>

Commission on Higher Education [CHED]. (2020, June 23). RECONSTITUTION OF THE CHED GENDER AND DEVELOPMENT (GAD) FOCAL COMMITTEE, AMENDING CSO NO. 90, SERIES 2014. CHED. <https://ched.gov.ph/wp-content/uploads/CSO-NO.-34-S.-2020.pdf>

Condron, C., Power, M., Mathew, M., Lucey, S., Henn, P., Dean, T., Kirrane-Scott, M., Eppich, W., & Lucey, S. M. (2024).

Gender Equality Training for Students in Higher Education: A Scoping Review. (Preprint). *JMIR Medical Education*.
<https://doi.org/10.2196/60061>

Corneliussen, H. G., Piscarella, A., & Castañeda-Navarrete, J. (2023).
Gender, digital transformation and AI.
ResearchGate.
<https://doi.org/10.13140/RG.2.2.19815.36003>

Duma, M. G. (2022).
GENDER MAINSTREAMING PLAN FOR GENDER-
RESPONSIVE PUBLIC SCHOOLS. *PUPIL*
International Journal of Teaching
Education and Learning, 5(3), 144-161.
<https://doi.org/10.20319/pijtel.2022.53.144161>

Devuser. (2023, July 18). *Records Management in the age of Big Data: Overcoming the challenges of information Overload*. Augusta Data Storage.
<https://www.augustadatastorage.com/records-management-in-the-age-of-big-data-overcoming-the-challenges-of-information-overload/>

Dr. Shamia WynnDr. John R. Kuhn, Jr. of
Liberty University (2021) The Financial
Impact of Manual Inventory Record
Errors
https://ijbssnet.com/journals/Vol_12_No_10_October_2021/2.pdf?

Garza, M. Y., Williams, T., Ounpraseuth, S.,
Hu, Z., Lee, J., Snowden, J., Walden,
A. C., Simon, A. E., Devlin, L. A.,
Young, L. W., & Zozus, M. N. (2023).
Error Rates of data processing Methods
in Clinical Research: A Systematic
Review and Meta-Analysis of Manuscripts
identified through PubMed. Research
Square (Research
Square).[https://pmc.ncbi.nlm.nih.gov/ar
ticles/PMC10775420/](https://pmc.ncbi.nlm.nih.gov/articles/PMC10775420/)

Gender and Development Corner | Isabela
State University. (n.d.).
[https://isu.edu.ph/gender-and-
development-corner/](https://isu.edu.ph/gender-and-development-corner/)

GENDER - CGIAR. (2021, August 3). CGIAR.
[https://www.cgiar.org/annual-
report/performance-report-2020/gender/](https://www.cgiar.org/annual-report/performance-report-2020/gender/)

Gholami, M., Kintzer, E., Wong, M., Smith, D., & Depp, C. (2025).

Implementation and impact of a diversity supplement Repository.

Journal of Clinical and Translational Science, 1-17.

<https://doi.org/10.1017/cts.2025.70>

Gillis, A. S. (2024, August 9). What is the software development lifecycle (SDLC)? *Search Software Quality*.

<https://www.techtarget.com/searchsoftwarequality/definition/software-development-life-cycle-SDLC>

Guido. (2025, April 29). *Gender equality and development in Asia and the Pacific*.

Asian Development Bank.

<https://www.adb.org/what-we-do/topics/gender>

Gupta, G. R., Grown, C., Fewer, S., Gupta, R., & Nowrojee, S. (2023).

Beyond gender mainstreaming:
transforming humanitarian action,
organizations and culture. *Journal of
International Humanitarian Action*,
8(1). <https://doi.org/10.1186/s41018-023-00138-1>

Handayani, F., et al. (2021). *Quality Assurance of Sayurbox Mobile Application Using ISO/IEC 25010*. IRJAES conference/article (2021).
https://irjaes.com/wp-content/uploads/2021/08/IRJAES-V6N3P148Y21.pdf?utm_source

Haas, O., Hutzler, M., Egginger, T. H., Maier, A., & Rothgang, E. (2021). Automating Time-Consuming and Error-Prone manual nursing management documentation processes. *CIN Computers Informatics Nursing*, 39(10), 584-591.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8505160/>

IBM. (2024, December 13). Software Development Life Cycle (SDLC). IBM.
<https://www.ibm.com/think/topics/sdlc?>

Indeed Editorial Team. (2025, June 6). *What is records management? (With benefits and life cycle)*. Indeed Career Guide. <https://www.indeed.com/career-advice/career-development/records-management>

The impact of records management on the effectiveness of human resource management in the public sector. (2024, June 7).

International Journal of Research and Innovation in Social Science.
<https://rsisinternational.org/journals/ijriss/articles/the-impact-of-records-management-on-the-effectiveness-of-human-resource-management-in-the-public-sector/>

Jane Wanjiru Njuki, Geoffrey Muchiri Muketha and John Gichuki Ndia (September 2022).

AN EXTENDED SECURITY MEASUREMENT
FRAMEWORK FOR OPEN-SOURCE ENTERPRISE
RESOURCE PLANNING SOFTWARE SECURITY.
https://aircconline.com/ijnsa/V14N5/14522ijnsa05.pdf?utm_source

Januhari, N. N. U., Setyanto, A., Kusrini, K., Utami, E., & Martínez-Béjar, R. (2025).

Evaluation of E-Learning Usability
Based on ISO 25010 with Hofstede's
Cultural Dimensions as Moderation: A
PLS-SEM Study in Higher Education.
*Applied Information System and
Management (AISM)*, 8(1), 159-168.
<https://doi.org/10.15408/aism.v8i1.4573>
8

Javier, B. S., Paliuanan, L. P., Talamayan, C. T., Agpalza, J. K. A., & Agoto, J. S. (2022).

*Development and Evaluation of the
Institutionally Farmed Research On-line
Repository and Management System
(InFORMs) towards Knowledge-Sharing and
Utilization.* arXiv.org.
<https://doi.org/10.22937/IJCSNS.2022.22.5.99>

Karima Moumane (2024) ISO/IEC 25010-based
Quality Evaluation of Three Mobile

Applications for Reproductive Health
Services in Morocco.

https://pdfs.semanticscholar.org/faef/96fb310aaa0dba5a93b246b4c3b10ea6019b.pdf?utm_source

Keibach, E., & Shayesteh, H. (2022). BIM for Landscape Design Improving Climate Adaptation Planning: The Evaluation of Software Tools Based on the ISO 25010 Standard. *Applied Sciences*, 12(2), 739.
<https://doi.org/10.3390/app12020739>

Lusiani, S., & Princes, E. (2024).
Evaluating the Effectiveness of Mobile JKN Application in Indonesia: A User-Centric Approach Using the ISO 25010 Quality Model. *Journal of Logistics, Informatics and Service Science*, 11(10), 485-500.
<https://www.aasmr.org/liss/Vol.11/No.10/Vol.11.No.10.28.pdf>

Maciel, R. S. P., Valle, P. H. D., Santos, K. S., & Nakagawa, E. Y. (2024).
Systems Interoperability Types: A tertiary study. *ACM Computing Surveys*,

56(10), 1-37.

<https://doi.org/10.1145/3659098>

Malak, H. A. (2025, February 11). *What is Data Automation? Why Is It Important?* Information Management Simplified.
<https://theecmconsultant.com/data-automation/>

Maxwell, J. A. (2022). Designing integrative mixed methods research. In *Elsevier eBooks* (pp. 441-449).
<https://doi.org/10.1016/b978-0-12-818630-5.11043-7>

Mbabe, W., Ajayi, O., Bagula, A., & Schoeman, N. (2021). A workflow system for managing ethical clearance in research work. *ResearchGate*.
https://www.researchgate.net/publication/356428947_A_Workflow_System_for_Managing_Ethical_Clearance_in_Research_Work

Mena, D., & Santorum, M. (2020). Maintainability and portability evaluation of the REACT Native Framework applying the ISO/IEC 25010. In *Advances in intelligent systems and*

computing (pp. 429-439).
https://doi.org/10.1007/978-3-030-59194-6_35

MOEYS Gender Mainstreaming Plan for Education 2021-2025. (2023, July 7).
<https://nepcambodia.org/2023/07/07/moey-s-gender-mainstreaming-plan-for-education-2021-2025/>

Moreno, F. (2023). *Gender-responsive budgeting in the Philippines: Comparative review of best practices and challenges*.

International Technology Management Corp. Munich Personal RePEc Archive.
https://mpira.ub.uni-muenchen.de/117899/1/__Gender-responsive%20Budgeting%20-%20April%2025%2C%202023%20-%20FINAL.pdf?utm_source

Najjar, Y. (2024, September 18). *The significance of user training in IT support*.

<https://www.linkedin.com/pulse/significance-user-training-support-yaser-najjar-vgntf>

O'Neill, V., & Soh, B. (2022). Improving fault tolerance and reliability of heterogeneous Multi-Agent IoT systems using intelligence transfer. *Electronics*, 11(17), 2724.
<https://doi.org/10.3390/electronics11172724>

Ose, S. O., Karlstrøm, H., Sand, K., & Lassemo, E. (2025). Has mainstreaming gender in research policy increased research with a gender perspective? *OUP Academic*.
<https://doi.org/10.1093/reseval/rvaf014>

Palma, E. J. C. (2024). *Gender and development implementation in public higher education institutions:*

Basis for a capability enhancement framework. Northern Iloilo State University. <https://www.apjaet.com/wp->

content/uploads/2024/09/Gender-and-Development-Implementation-in-Public-Higher-Education-.pdf?utm_source

Palma, E. J. C. (2024). *Gender and development implementation in public higher education institutions:*

A basis for a capability enhancement framework. Science Gate.

https://www.science-gate.com/IJAAS/Articles/2025/2025-12-02/1021833ijaas202502011.pdf?utm_source

Panorama Consulting Group. (2024, August 21). Navigating cultural resistance in IT transformation [Using Change Management]. *Panorama Consulting Group.*

<https://www.panorama-consulting.com/navigating-cultural-resistance-in-it-transformation-using-change-management/>

Perez, D., Velicaria, A., Berja, V. R., Cruz, M., & Palmiano, D. A. (2023).

Pathways and roadblocks in gender and development: A systematic review of institutional practices in higher education.
https://www.ijfmr.com/papers/2025/4/53467.pdf?utm_source

Philippine Commission on Women. (2022).

Updated Gender Equality and Women's Empowerment (GEWE) Plan 2019-2025.
Philippine Commission on Women.
https://www.pnri.dost.gov.ph/images/GADCorner/PCW-Updated-Gender-Equality-and-Womens-Empowerment-Plan-2019-2025-2022.pdf?utm_source

Philippine Statistics Authority. (2021).

Approval and adoption of the updated Philippine core gender and development indicators (PSA Board Resolution No. 08, Series of 2021).
<https://psa.gov.ph/system/files/psa->

[board/PSA%2520BOARD%2520RESO%252008%25202021.pdf?utm_source=](https://www.psa-board.org/PSA%2520BOARD%2520RESO%252008%25202021.pdf?utm_source=)

Puspaningrum, A. S., Rochimah, S., & Akbar, R. J. (2017). Functional Suitability Measurement using Goal-Oriented Approach based on ISO/IEC 25010 for Academics Information System. *Journal of Information Systems Engineering and Business Intelligence*, 3(2), 68.
<https://doi.org/10.20473/jisebi.3.2.68-74>

Quickbase, I. (2024, June 5). Benefits of Cloud-Based Document Management Systems. *Quickbase*.
<https://www.quickbase.com/blog/benefits-of-cloud-based-document-management-systems>

Rañon, D. N. (2024). *Implementation level of gender mainstreaming in one state college in Camarines Sur, Philippines:*

An analysis using the Gender Mainstreaming Evaluation Framework (GMEF). Camarines Sur Polytechnic Colleges.

https://www.ijfmr.com/papers/2024/6/30594.pdf?utm_source

Ray Waldron, (2024, June 5).

Benefits of Cloud-Based Document Management Systems. Quickbase.

<https://www.quickbase.com/blog/benefits-of-cloud-based-document-management-systems>

ResearchFDI. (2024, December 13). How technology is changing the landscape of economic development. *ResearchFDI*.

<https://researchfdi.com/how-technology-is-changing-the-landscape-of-economic-development/>

Record Management System with Document Control. (n.d.).

<https://ijmra.in/v4i12/23.php>

Sarwosri, S., Rochimah, S., Yuhana, U. L., & Hidayat, S. B. (2023). Software quality measurement for functional suitability, performance efficiency, and reliability characteristics using analytical hierarchy process. *JOIV International Journal on Informatics Visualization*,

7(4), 2421.

<https://doi.org/10.62527/joiv.7.4.2441>

Shahrzadi, L., Mansouri, A., Alavi, M., & Shabani, A. (2024). Causes, consequences, and strategies to deal with information overload: A scoping review. *International Journal of Information Management Data Insights*, 4(2), 100261.
<https://doi.org/10.1016/j.jjime.2024.100261>

Shahid, M. A., Alam, M. M., & Su'ud, M. M. (2023). Achieving reliability in cloud computing by a novel hybrid approach. *Sensors*, 23(4), 1965.
<https://doi.org/10.3390/s23041965>

Singh, S., Rahul, K., Paliwal, M., Wani, I. A., & Suri, S. (2025). Gendering the digital divide: a systematic review of women's digital inclusion challenges and emerging research directions. *Digital Transformation and Society*, 1-29.

<https://doi.org/10.1108/dts-04-2025-0083>

Skan.Ai. (2025, June 9). 5 Key benefits of Data-Driven Decision making. *Skan AI*.
<https://www.skan.ai/blogs/5-key-benefits-of-data-driven-decision-making>

Slyngstad, L., Nylenna, M., Nytrø, Ø., & Gullslett, M. K. (2022). How do different health record systems affect home health care? A cross-sectional study. *International Journal of Environmental Research and Public Health*, 19(5), 2800.
<https://www.mdpi.com/1660-4601/19/5/2800>

The National Gender and Development Resource Program | Philippine Commission on Women. (n.d.). <https://pcw.gov.ph/the-national-gender-and-resource-program/>

The impact of records management on the effectiveness of human resource management in the public sector. (2024, June 7). *International Journal of Research and Innovation in Social*

Science.

<https://rsisinternational.org/journals/ijriss/articles/the-impact-of-records-management-on-the-effectiveness-of-human-resource-management-in-the-public-sector/>

Turkish Journal of Computer and Mathematics Education. (2021). Errors due to manual documentation reduce reliability of the system. \ Turkish Journal of Computer and Mathematics Education, 12(11), 6371-6380.

<https://turcomat.org/index.php/turkbilm/et/article>

Turner, T. (2025, January 28). Building an Inclusive America - a framework for LGBTQ+ rights. *Freedom For All Americans*.

<https://freedomforallamericans.org/framework-for-lgbtq-rights/>

UNESCO. (2025, March 30). Gender mainstreaming in higher education: Global trends and perspectives.

<https://www.unesco.org/en/gender-equality/>

Usman, M., Felderer, M., Unterkalmsteiner, M., Klotins, E., Mendez, D., & Alégroth, E. (2020). Compliance Requirements in Large-Scale Software Development: an industrial Case study. In Lecture notes in computer science (pp. 385-401).
<https://arxiv.org/abs/2103.01821?>

Vidjikan, S., & Neville, M. (2025, February 11). Why use PHP in 2025? Advantages and Disadvantages. *Softjournal Inc.*
<https://softjournal.com/insights/pros-and-cons-of-php-programming-language>

What is Gender and Development (GAD)? | Gender and Development - Commission on Audit. (n.d.).
<https://web.coa.gov.ph/gad/index.php/docs/what-is-gender-and-development-gad/>

What Is Records Management? (With Benefits and Life Cycle). (2024, August 16). *Indeed.* <https://www.indeed.com/career-advice/career-development/records-management?fbclid=IwY2xjawIU->

[rFleHRuA2FlbQIxMAABHSURNjZ72j3cr4WUUK5q2hSudGdRA4V7awUnqI0fou72jfz1CvnEMOeHzQaem 4FEoQF5glEXU1CP3IRnr0g](https://arxiv.org/abs/2406.14489)

Yilmaz, R., & Buzluca, F. (2024, June 20). A *fuzzy Logic-Based quality model for identifying microservices with low maintainability*. arXiv.org.
https://arxiv.org/abs/2406.14489?utm_source

Yin, A. L., Guo, W. L., Sholle, E. T., Rajan, M., Alshak, M. N., Choi, J. J., Goyal, P., Jabri, A., Li, H. A., Pinheiro, L. C., Wehmeyer, G. T., Weiner, M., Safford, M. M., Campion, T. R., & Cole, C. L. (2021). Comparing automated vs. manual data collection for COVID-specific medications from electronic health records. *International Journal of Medical Informatics*, 157, 104622.
https://www.researchgate.net/publication/355507766_Comparing_Automated_vs_Manual_Data_Collection_for_COVID-Specific_Medications_from_Electronic_Health_Records

Zai, G. Y. (2023, October 9). The disadvantages of not having an attractive and User-Friendly website. <https://www.linkedin.com/pulse/disadvantages-having-attractive-user-friendly-website-ghous-yousuf>

Zulfa, F., Munawaroh, H., & Rochimah, S. (2023). Portability Characteristics Evaluation of MyITS Mobile using ISO/IEC 25010 Quality Standard. *2020 International Seminar on Application for Technology of Information and Communication (iSemantic)*. <https://doi.org/10.1109/isemantic50169.2020.9234241>